

RIZVI NCC UNIT MANAGEMENT WEBSITE

A Project Report

Submitted in Partial fulfilment of the

Requirements for the award of the Degree of

BACHELOR OF SCIENCE (COMPUTER SCIENCE)

By

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RIZVI COLLEGE OF ARTS, SCIENCE AND COMMERCE

(Affiliated to University of Mumbai)

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2025-2026

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CERTIFICATE

This is to certify that the project entitled, “**RIZVI NCC UNIT MANAGEMENT WEBSITE**”, is benefited work of **SAYYED MOHAMMAD HANZALA MOHAMMAD YASIN**. Bearing Seat No.: Roll No. 64 submitted in Partial fulfilment of the requirements for the award of **Degree of BACHELOR OF SCIENCE in COMPUTER SCIENCE** from **University of Mumbai**.

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College Seal

ACKNOWLEDGEMENT

I owe special thanks to the Department of Computer Science of **Rizvi College of Arts Science and Commerce** for giving me a chance to prepare this project. I thank to the Principle, **Dr. S. ARUNACHALAM** for his leadership and management. I thank the Coordinator and Head of the Department **Professor Arif Patel** for providing us the required facilities and guidance throughout the course which culminated into this thesis. Last but not the least to the project guide for this odd semester – **Professor Javed Pathan** and also to my **Parents and Friends** for supporting me throughout the semester and helping me in finalizing the project. Also deep gratitude to the **Staff and Faculty of Rizvi College of Arts Science and Commerce** for their help and support.

**"RIZVI NCC UNIT MANAGEMENT WEBSITE" USING PYTHON,
HTML, AND CSS**

**RIZVI COLLEGE OF ARTS, SCIENCE AND
COMMERCE**
(Affiliated to University of Mumbai) MUMBAI-
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DECLARATION

I , **SAYYED MOHAMMAD HANZALA MOHAMMAD YASIN** , Roll No. **64**, hereby declare that the project synopsis entitle “**RIZVI NCC UNIT MANAGEMENT WEBSITE**”, submitted for approval , **for Bachelors of Science in Computer Science SEM V** project. For Academic Year **2025-2026**

Signature of the Guide

Signature of the Student

Place:

INTRODUCTION:

The Rizvi NCC Unit Management Website is a platform created to make the management of our NCC unit simple, fast, and well-organized. In today's time, almost everything is shifting towards digital systems, so this website is developed to manage cadets and NCC activities online instead of using old manual methods. This helps in saving time, avoiding mistakes, and improving the overall working of the unit.

In an NCC unit, there are many tasks that need to be done regularly such as cadet registration, attendance management, event scheduling, announcements, notices, and performance tracking. Doing all this on paper is not only slow but also difficult to maintain for a long time. To solve this problem, the Rizvi NCC Unit Management Website provides a single platform where all these tasks can be completed in an easy and systematic way.

Cadets can use this website to register themselves, update their details, check their attendance, and get information about upcoming events and parades. At the same time, ANOs and admin staff can manage cadet records, add important updates, upload circulars, and monitor the overall performance of cadets. This makes the communication between cadets and officers smooth and effective.

The website is designed to be simple, responsive, and user-friendly. It can be used both on mobile and desktop, so cadets and officers can access it anytime and from anywhere. It also has a secure login system to make sure that only authorized users can access the data. This keeps cadet records safe and private.

Apart from management, the website also represents the true spirit of NCC, which is Discipline, Unity, and Leadership. It not only helps in handling daily tasks but also motivates cadets to be more responsible, active, and engaged in unit activities.

OBJECTIVES:

1. The main objective of this website is to make the management of NCC activities easy and digital instead of using manual work.
2. It provides a simple and user friendly platform for cadets, ANOs and admins to manage their tasks properly.
3. It helps in accurate cadet registration, attendance record and performance tracking.
4. The website keeps all the cadet data safe and secure without the risk of losing it.
5. It improves communication between cadets and officers by showing notices, updates and announcements in one place.
6. It motivates cadets to take part in parades, camps and other NCC activities by keeping them informed.
7. It saves time and reduces errors which usually happen in manual record keeping.
8. It spreads the core values of NCC like discipline, unity and leadership through a modern digital platform.
9. The website can be accessed from mobile or computer so it is easy to use anytime and anywhere.
10. The aim is to make the Rizvi NCC unit more organized, transparent and efficient with the help of technology.

SCOPE:

1. The project involves the development and design of an official website for the Rizvi NCC Unit.
2. It includes creating a simple and user friendly interface with relevant sections and pages that are easy to use.
3. The website will be responsive so that it can be accessed on mobile phones as well as computers without any problem.
4. The structure of the website will ensure easy navigation and quick access to important information like notices, events, and attendance.
5. The website will provide separate logins for cadets, ANOs, and admins to manage their tasks properly.
6. It will allow cadets to register online, check their attendance, and stay updated with unit activities.
7. It will also allow ANOs and admins to manage cadet records, upload announcements, and monitor cadet performance.
8. The system will save time, reduce paperwork, and make the overall management of the NCC unit more accurate and transparent.

By defining the scope of the project, the Rizvi NCC Unit Management Website can be developed and implemented effectively for the benefit of both cadets and officers.

METHODOLOGY:

1. Planning and Requirement Gathering

In the first step, we identified the need for a Breast Cancer Detector website and collected all the necessary requirements. The main focus was on providing a user-friendly website that can detect breast tumors and give accurate results.

2. Template Design

After planning, we created a simple template to visualize the structure of the website. This gave us a clear idea of how the website will look and what sections or pages should be included.

3. Front Page Development

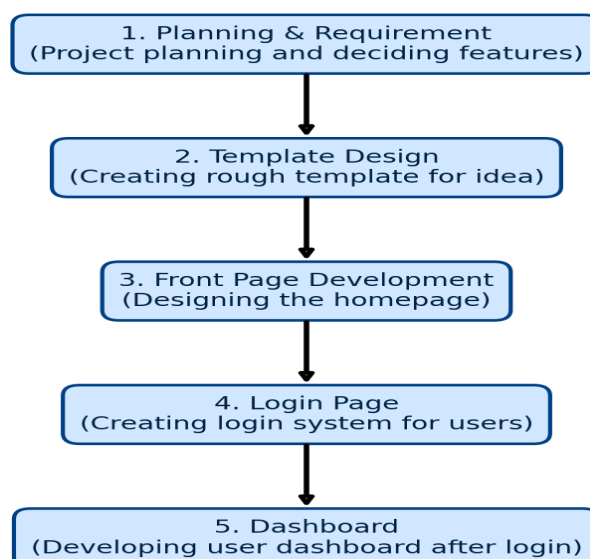
The next step was to design and build the front page of the website. The front page acts as the entry point for the users, providing easy navigation and access to different sections like login, awareness information, and diagnosis.

4. Login Page Development

A secure login page was created where users can enter their credentials to access personalized features. This ensures data privacy and prevents unauthorized access.

5. Dashboard Creation

Once logged in, users are redirected to a dashboard. The dashboard provides different services like uploading sonography images, viewing results of diagnosis, and accessing awareness material. It acts as the main working area of the website.



TOOLS AND TECHONOLOGIES:

1. The project uses different tools and technologies to make the website easy, useful and reliable for both users and admins.
2. The front-end of the website is developed using HTML and CSS. This makes the website user-friendly, attractive and responsive so that users can easily access it.
3. The back-end of the website is built using Django framework of Python. Django helps in handling the logic, storing and managing the data, and connecting the front-end with the database.
4. The authentication system is created using Django's in-built authentication, so that users and admins can log in securely.
5. For the future, if the project works successfully, the website can also be hosted online so that more people can access it easily from anywhere.

* The above mentioned are the tools and technologies which are required and used by the developer.

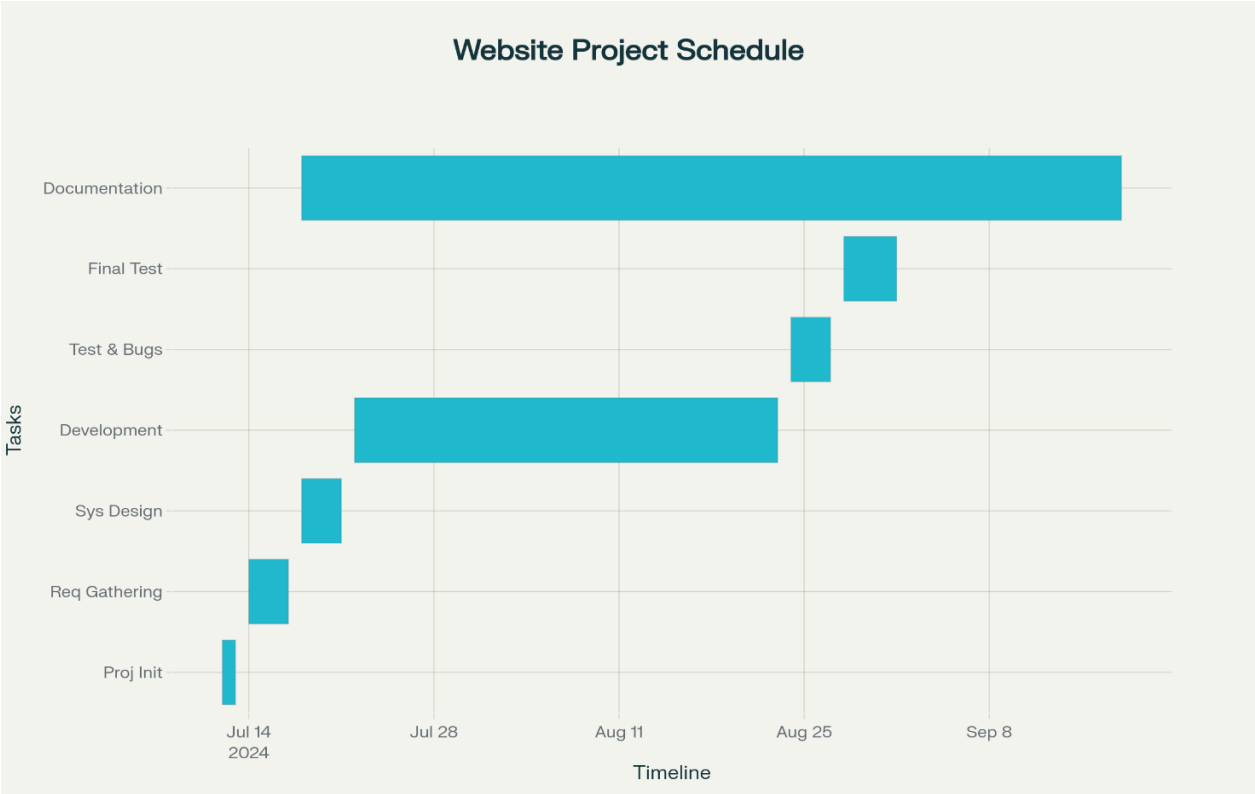
* There are a few basic requirements mentioned below for both the User as well as for the developer.

1. The requirements for the developer are a laptop or desktop with good specifications and internet connection to run Django smoothly.
2. The requirements for the users are a mobile phone or computer with a good browser and internet connection to access the website.

TIMELINE:

The timeline of the overall project can be represented with the Gantt chart below.

GANTT CHART



Task	Start date	End Date	Duration(Days)
Project Initiation	12-07-2024	13-07-2024	1
Requirement Gathering	14-07-2024	17-07-2024	3
System Design	18-07-2024	21-07-2024	3
Development	22-07-2024	23-08-2024	32
Testing and Bug fixes	24-08-2024	27-08-2024	3
Final Testing	28-08-2024	01-09-2024	4
Documentation	02-07-2024	18-09-2024	64

EXPECTED OUTCOMES:

1. Enhanced Information Access:

The website will allow cadets, staff members, and other visitors to easily access up-to-date information about NCC units, cadet details, training schedules, and activities. This ensures transparency and makes processes more convenient for everyone.

2. Spreading Awareness:

The website will showcase NCC activities, training programs, and achievements using text, images, and videos. This will help increase awareness about the benefits and importance of NCC among students and the community.

3. Transparent and Accessible Information:

The system will make NCC data and processes clear and accessible. A simple and user-friendly interface will help users quickly find the information they need, building trust and confidence in the website.

4. Continuous Improvement and Adaptation:

The website will be regularly updated based on user feedback. Suggestions and reports from users will help the development team improve features and address any issues, making the website better over time.

5. Security Measures:

Strong security will protect the personal and official data of all NCC members. The website will use secure authentication, encrypted databases, and role-based access to keep all information safe.

6. User-Friendly Interface:

The website will be responsive and easy to use on any device. Cadets, staff, and administrators can access data efficiently, making the overall experience quick and effortless.

The NCC Management website will be a dynamic online platform that empowers cadets and staff with accurate information, promotes awareness of NCC activities, and creates a safe, transparent, and healthy digital environment for all users

ADVANTAGES:

1. Efficient Information Management:

The website enables quick and organized storage of cadet profiles, attendance records, training schedules, and activity reports. Both administrators and cadets can easily find, add, or update relevant information, which streamlines communication and record-keeping for the whole NCC unit.

2. Integrated NCC Activities:

Event calendars, notifications, and resource sharing features keep users updated about upcoming training camps, drills, competitions, and awareness drives. This helps in better planning, higher participation rates, and improved coordination among cadets and staff.

3. User-Friendly and Accessible:

The platform has a responsive layout, making it easy for anyone cadet, officer, or teacher to log in and use on computers, tablets, or phones. User-friendly navigation and service guides boost adoption and reduce training time for new members.

4. Digital Documentation and Record Keeping:

All NCC activities, certificates, and attendance are stored digitally, reducing paperwork and making future audits faster and more reliable. Past events, achievements, and certificates can be accessed and verified by authorized users anytime, anywhere.

5. Supports Leadership and Teamwork Development:

Sections for reporting, feedback, and discussion forums are included to help cadets share experiences and ideas. Online community features can promote teamwork and confidence-building among NCC members, reinforcing the organization's mission.

6. Promotes Social Awareness and Patriotism:

Multimedia, blogs, and photo galleries showcase achievements, promote community campaigns, and share educational content about NCC, safety, and national integration. This helps boost awareness both inside and outside the institution

LIMITATIONS:

1. Limited Bulk Data Entry:

The website currently manages cadet entries, attendance, and reports one at a time. Bulk uploads and automatic integrations are not present, which might be less efficient for very large NCC units.

2. Dependent on Internet Access:

All features registration, updating, viewing require a stable internet connection and a browser-enabled device. During internet outages or in remote areas, access to website features may be restricted.

3. No Real-Time Human Support:

For personalized questions, clarifications, or problem-solving, users must rely on digital contact forms or support tickets. There is no instant chat or phone-based support built into the platform.

4. Accuracy Bound by User Input:

Cadet achievements, attendance, and activity records rely on data entered by users or staff. Mistakes and omissions in data entry can affect the accuracy of reports and certificates.

5. Security and Privacy:

Although strong security measures are applied, risks such as unauthorized access or data breaches cannot be completely ruled out. Regular updates and vigilance are required.

REQUIREMENTS:

1. Device and Connectivity:

To use the website, each user needs a device (computer, laptop, tablet, or mobile phone) with internet connectivity and a modern web browser.

2. Technical Setup for Development:

Developers require a laptop or PC with Windows, Mac, or Linux OS, and a steady internet connection. Familiarity with website frameworks (such as Django), and installation of relevant Python libraries are needed.

3. Software Installations:

The server-side Python environment must have all required libraries installed before project setup. These may include Django, database drivers, and framework-specific modules for proper functioning and security.

4. User Training:

Basic guidance in using the website, uploading cadet data, generating reports, and handling event modules is beneficial for both new cadets and staff member

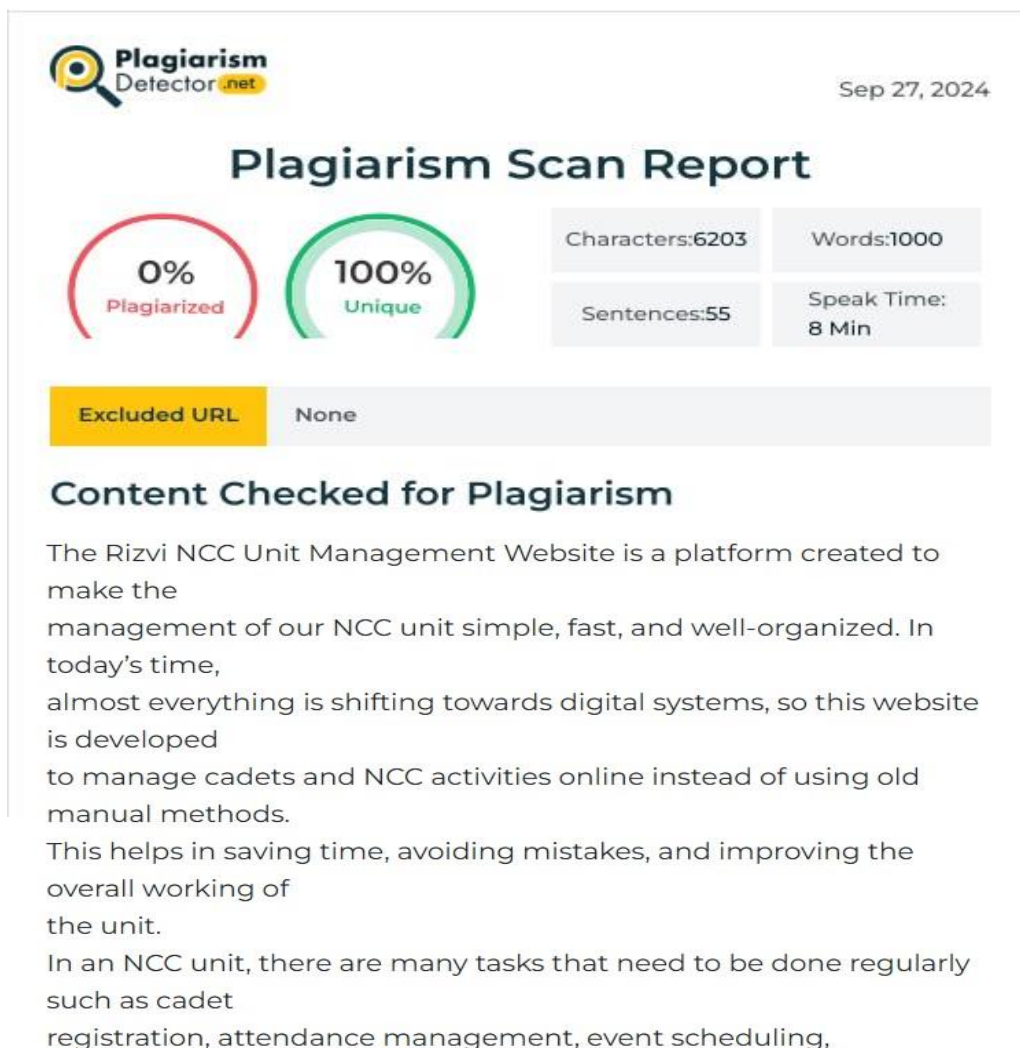
REFERENCE:

- 1.Introduction: Details about the NCC aims, training, and impact can be found in government and institutional handbooks and reputable educational websites.
- 2.Objectives: Review recognized university and NCC official resources for mission, vision, and objectives.
- 3.Gantt Chart: For project planning references, use official YouTube tutorials and PM guides.
4. Technology Details: See official Django documentation and Python library instructions for setup guidance.

PLAGIARISM REPORT

A plagiarism report is a document or a summary that provides information about the presence of plagiarism in a piece of written or academic work. Plagiarism refers to the act of using someone else's words, ideas, or work without proper attribution or permission, presenting them as your own. Plagiarism is considered unethical and can have serious consequences, particularly in academic and professional settings.

A plagiarism report is typically generated by plagiarism detection software or services. It scans a given document or text for similarities to existing sources, such as published articles, books, websites, and other written material. When the software identifies matching or highly similar content, it highlights or marks the specific passages that may be co-related.



DECLARATION

I hereby declare that the project entitled, “**RIZVI NCC UNIT MANAGEMENT WEBSITE** ” done at **Rizvi College of Arts, Science and Commerce** , has not been in any case duplicated to submit to any other university for the award of any degree. To the best of my knowledge other than me, no one has submitted to any other university. The project is done in partial fulfilment of the requirements for the award of degree of **BACHELOR OF SCIENCE (COMPUTER SCIENCE)** to be submitted as final semester project as part of our curriculum.

**SAYYED MOHAMMAD HANZALA
MOHAMMAD YASIN**

ABSTRACT

The NCC Unit Management Website is a simple and effective online platform designed to help manage the day-to-day activities of NCC cadets and staff. This website provides a centralized system where cadets can register, update their personal information, and keep track of their attendance and participation in various NCC events and training programs. It also allows instructors and administrators to easily manage cadet records, training schedules, and announcements.

Built using Django and Python, the website ensures data security by providing login access based on user roles such as cadet, instructor, and admin. This role-based access helps protect sensitive information and ensures that users can only access data relevant to them. The user-friendly interface makes it easy for all users to navigate the website, even if they are not tech-savvy.

The shift from manual, paper-based record keeping to this digital system brings many benefits. It saves time, reduces errors, and makes updating and viewing cadet information much faster. Training reports, attendance records, and event details are stored digitally, making it easier to access and review whenever needed.

In addition to managing records, the website also supports communication within the NCC unit by providing features such as announcements, event calendars, and feedback forms. This helps improve coordination among cadets, instructors, and officers and promotes teamwork and discipline.

Overall, this website plays a significant role in organizing NCC operations more efficiently and transparently. It supports the growth and development of cadets by providing them with easy access to their records and keeping the entire unit informed and connected.

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CHAPTER 1. INTRODUCTION

1.1 Introduction to the Website

The Rizvi NCC Unit Management Website is a dedicated online platform developed to modernize and streamline the administration and activities of the National Cadet Corps (NCC) unit at Rizvi College of Arts, Science and Commerce. The NCC unit at Rizvi plays a vital role in nurturing qualities of leadership, discipline, teamwork, and national responsibility among students, while also encouraging their participation in various national and state-level camps, social initiatives, and sporting competitions.

Traditionally, NCC management tasks such as cadet registration, attendance tracking, achievement recording, and event scheduling were handled manually using paper registers or basic spreadsheets. This cumbersome process was often prone to data loss, duplication, errors, and significant delays in communication. There was also a risk that valuable information about cadet achievements or event participation could be missed or misplaced during transitions from one academic year to the next.

The Rizvi NCC Unit Management Website has been designed to solve these challenges by providing a secure, user-friendly, and centralized web-based system. All core activities including new cadet enrollments, personal data updates, activity and training records, announcements, attendance, and achievement logs are now digital and accessible in real time. Cadets can easily check their progress, view event schedules, and receive information about upcoming training or competitions at any time. Instructors and administrators gain powerful tools for efficiently managing individual records, sharing news, issuing certificates, and maintaining accurate databases for internal reporting as well as for university or unit audits.

This innovative website not only improves day-to-day operations but also promotes transparency, communication, and motivation within the NCC unit. By digitalizing essential processes, the platform empowers every member to focus more on holistic development whether it's excelling in national camps like the Thal Sainik Camp or Republic Day Parade, or contributing to community and campus initiatives.

Ultimately, the Rizvi NCC Unit Management Website signals a significant advancement in the college's approach to youth leadership and institutional management, making it easier for cadets and faculty to coordinate, collaborate, and celebrate their successes as a team.

1.2 Problem Definition

Before using this website, managing the NCC unit involved many problems. Most of the work like registering new cadets, keeping track of their attendance, updating their details, and organizing training events was done manually on paper or simple spreadsheets. This made the process slow and confusing at times. Records could get lost, duplicated, or incomplete. It was difficult for instructors and administrators to keep everything organized and up to date.

Communication between cadets and staff was sometimes delayed because there was no easy way to share announcements, schedules, or results quickly. Many important activities like tracking achievements or planning events were hard to follow in real time. These problems caused delays and made it harder to keep all cadets engaged and informed.

The Rizvi NCC Unit Management Website solves these issues by providing a simple and reliable online platform. It automates registration, attendance, event scheduling, and record keeping. It also improves communication by sending announcements and updates instantly to all users. This makes managing the NCC unit faster, more accurate, and more transparent. Everyone from cadets to officers can easily access the information they need anytime, anywhere.

1.3 Aim

The aim of the Rizvi NCC Unit Management Website is to make managing the NCC unit easier, faster, and more organized. The website provides a simple and easy way for cadets and staff to handle tasks like registration, attendance, event scheduling, and record keeping. It helps to reduce mistakes and saves time by digitizing all important information in one place.

The website is designed with a user-friendly interface so that all users, whether they are cadets, instructors, or administrators, can use it without difficulty. It ensures that everyone has quick access to their data and important updates related to NCC activities.

The main goal is to improve the overall efficiency and communication of the NCC unit, making sure that the unit runs smoothly and that cadets get the full benefit of their training and activities.

1.4 Objectives

To make the management of NCC cadets easy, accurate, and quick using a digital platform.

To keep good records of cadet registration, attendance, and achievements with less mistakes.

To provide a simple and friendly website that all users can easily use without confusion.

To help instructors and administrators organize events, training schedules, and communication efficiently.

To make sure all cadets get updates and information about NCC activities on time. To support smooth coordination and better teamwork within the NCC unit.

1.5 Goal

The main goal of the Rizvi NCC Unit Management Website is to make managing the NCC unit easy, quick, and error-free. The website aims to provide a simple platform where cadets, instructors, and staff can handle all NCC-related activities in one place, such as registration, attendance, events, and record keeping.

It also aims to improve communication between cadets and officers by sharing important news, schedules, and announcements quickly and clearly. By making these tasks digital and accessible online, the website helps the NCC unit run smoothly and keeps all members informed.

Ultimately, the goal is to support cadet training, discipline, teamwork, and leadership development by making administrative work simpler and faster. This will help cadets focus on their growth and prepare better for their duties and future careers.

1.6 Need of the System

The Rizvi NCC Unit Management Website is needed to make handling all NCC-related work easier and faster. Before this, managing cadet information, attendance, training schedules, and events was done on paper or basic software, which was slow and could cause errors or lost information.

A digital website helps keep all this data in one safe place, making it easy to update and access anytime. It also helps improve communication by sharing announcements and event details quickly among cadets and staff.

This system reduces the workload of officers and organizers by automating many tasks, like attendance marking and report generation. It also helps cadets stay informed and motivated by providing quick access to their progress and upcoming activities.

CHAPTER 2. REQUIREMENT SPECIFICATION

2.1 Introduction

In developing the Rizvi NCC Unit Management Website, it is important to clearly define the requirement specifications. These are the detailed needs and features that the website must have to work well and meet the expectations of the users. The requirement specifications act as a guide for designing, building, and testing the website to make sure it serves all necessary functions properly.

This website aims to solve the common problems faced in managing NCC units, such as keeping accurate records of cadet information, attendance, events, and communication. The specifications ensure that the website is user-friendly, secure, and efficient so that cadets, instructors, and administrators can use it easily in their daily work.

By clearly outlining these requirements, developers, designers, and stakeholders can work together to create a reliable platform that supports the efficient management of the NCC unit. This will help the Rizvi NCC unit to be more organized, reduce manual errors, save time, and improve overall teamwork and communication.

2.2 System Environment

The system environment for the Rizvi NCC Unit Management Website includes the computers or servers hosting the website, the internet network allowing access via browsers on devices like laptops, tablets, or phones, and the software components such as Python, Django, and the database used to store and manage NCC data securely and efficiently. This environment ensures that the website runs smoothly, safely, and can be accessed easily by cadets, instructors, and administrators from any location.

2.3 Software Requirements

The software required for developing the Rizvi NCC Unit Management Website includes Visual Studio Code (VS Code) as the code editor, which provides useful features like code highlighting, debugging, and extensions to support Python and Django development. Python is the main programming language, and Django is the web framework used to structure the website and handle backend tasks. To keep project dependencies organized and isolated, setting up a virtual environment is recommended, ensuring smooth development and easy management of libraries.

2.4 Hardware Requirements

1. Processor (CPU): Intel Core i5 (6th generation or newer) or equivalent
2. Operating System: Microsoft Windows 10 Professional 64-bit
3. Memory (RAM): 8 to 16 GB or higher
4. Storage: Minimum 500 GB internal hard drive
5. Monitor/Display: 14-inch LCD with resolution 1600 x 900 or better
6. Network Adapter: Wireless adapter supporting 802.11ac 2.4/5 GHz for internet connectivity

2.5 Methodology

The development of the Rizvi NCC Unit Management Website will use the Spiral Model, which is a flexible and risk-focused software development approach. The Spiral Model works through multiple cycles, each including four main phases: Planning, Risk Analysis, Engineering, and Evaluation.

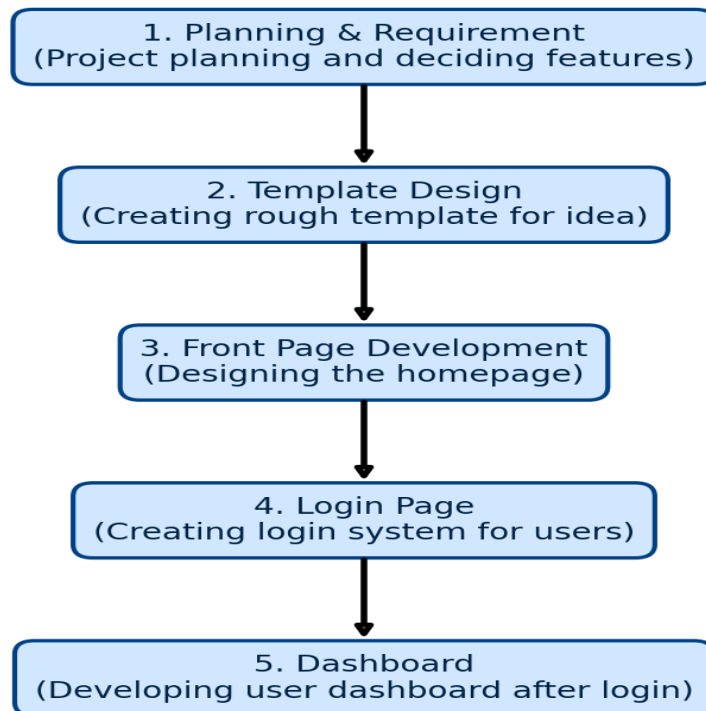
In the Planning phase, the project's goals, requirements, and limits are gathered and clearly defined. This helps set a clear path for development and ensures everyone's expectations are understood. During the Risk Analysis phase, the team identifies potential problems that could delay or affect the project, such as security risks or missing features. They then plan how to reduce or avoid these risks. The Engineering phase involves actual coding, building, and integrating planned features into the website. The team tests features and makes improvements based on feedback from previous cycles. Finally, in the Evaluation phase, the team collects feedback from users and tests the website thoroughly to make sure it meets quality standards and works as expected.

Each cycle improves on the previous one based on the evaluations and feedback, making the software more reliable and user-friendly with each iteration. This method is especially useful for projects like the NCC website, which may have evolving requirements and need careful risk management. It helps the development team focus on solving potential problems early, delivering a secure, well-functioning, and easy-to-use website that meets the needs of all users. This iterative process continues until the website is fully developed and ready to be deployed for everyday use.

2.6 Spiral Model

The Spiral Model is a software development approach that combines the strengths of both the iterative and waterfall models, providing a structured yet flexible way to manage complex projects. It involves repeated cycles called "spirals," where each spiral represents one iteration of development. Each cycle includes four important phases: Planning, Risk Analysis, Engineering, and Evaluation. In the Planning phase, project goals, requirements, and limits are defined clearly to set a solid foundation. During Risk Analysis, all possible risks such as technical challenges or unclear requirements are identified and solutions are planned to reduce those risks. The Engineering phase is where the actual building and testing of the software happens based on the plans and risk assessments. Finally, in the Evaluation phase, feedback is collected from users and stakeholders to check if the software meets their needs and quality standards. This feedback guides the planning of the next spiral. The process repeats with each spiral improving on the last one, allowing the project to adapt, reduce risks, and deliver a better product gradually. This model is especially useful for projects where requirements may change or are not completely understood from the start, as it allows early detection and handling of problems, better risk management, and continuous user involvement, leading to successful and reliable software delivery.

Diagram of Spiral model :



Why Spiral Model?

The Spiral Model is chosen for the Rizvi NCC Unit Management Website because it offers a powerful way to manage risks while allowing the project to adapt to changing needs. It combines the best features of iterative and traditional development methods, ensuring the project develops step-by-step with continuous feedback and improvement. This model begins with setting clear goals and analyzing potential risks, such as data loss or security issues, which helps avoid problems early. Each cycle through the phases of planning, risk analysis, engineering, and evaluation focuses on refining the project based on real feedback and any changing requirements. This flexibility is especially useful for complex

projects like the NCC website, where requirements may evolve during development. It also supports better communication with users, ensuring their feedback shapes the system. By using the Spiral Model, the development team can deliver a secure, functional, and easy-to-use website that meets the diverse and growing needs of the Rizvi NCC unit, reducing the chance of costly errors or project failure. Overall, it keeps the development process flexible, controlled, and focused on delivering the best product possible.

Applications of the Spiral Model

The Spiral Model is widely used in the software industry because it matches the natural way products develop, which is through learning and improving over time while keeping risks low for both customers and developers. Here are some key points about the Spiral Model:

1. Ideal for managing changing and evolving requirements in dynamic software projects.
2. Well-suited for developing complex systems where reducing risks is very important.
3. Effective in industries with strict rules and regulations where compliance is crucial.
4. Supports the iterative improvement of prototypes and early versions of software.
5. Valuable for research and development projects that involve experimentation and testing.
6. Helps in client-focused projects where active user involvement and feedback shape the development.

CHAPTER 3. SYSTEM ANALYSIS

3.1 System Analysis

System analysis is the process of studying and understanding how a system works by breaking it into smaller parts and looking at how these parts interact. It helps identify problems and find better ways to achieve the system's goals efficiently. In the case of the Rizvi NCC Unit Management Website, system analysis involves examining how cadets, instructors, and administrators currently manage registrations, attendance, events, and communications.

By analyzing these processes, we can find difficulties such as delays, errors, or lack of communication. System analysis helps gather requirements, understand user needs, and create models of how the new website should work. Its main goal is to bridge the gap between the current challenges and the desired improvements by proposing solutions that make the system faster, clearer, and easier to use. This will help the NCC unit operate smoothly, reduce manual work, and provide better service to all members

3.2 Analysis of System Without AI

Before designing and building the new NCC Unit Management Website, it is crucial to thoroughly understand the limitations and difficulties faced by the existing manual or non-automated systems. Currently, the system relies heavily on paper-based records and manual efforts from officers and staff for handling registrations, attendance, event planning, and communication. This dependence slows down processes significantly and increases the chance of human errors such as incorrect or missing data entries. These errors can cause confusion and affect the accuracy of cadet records and event management. Additionally, the manual workload is often overwhelming for personnel, leading to fatigue and decreased efficiency. Managing and organizing large volumes of cadet information, training schedules, and events becomes challenging without automated tools.

This results in delays in sharing important updates and announcements due to lack of instant communication methods. Another major concern is the difficulty in generating accurate, real-time reports, which impacts the decision-making and monitoring processes. By analyzing these pain points, the new digital NCC Unit Management Website can focus on automating these tasks, improving accuracy, speeding up communications, and making the overall management system more efficient and user-friendly for all members involved.

1. Human errors: Mistakes can happen due to manual data entry, leading to wrong or missing records.
2. Delayed processes: Manual updating slows down attendance, registration, and event management.
3. Heavy workload: Officers and staff managing everything by hand face fatigue and decreased efficiency.
4. Poor data management: Handling a large number of cadets and activities becomes difficult without automation.
5. Communication gaps: Lack of instant notifications causes delays in sharing important announcements.
6. Inaccurate reporting: Manual methods make it hard to get real-time and accurate reports for monitoring.

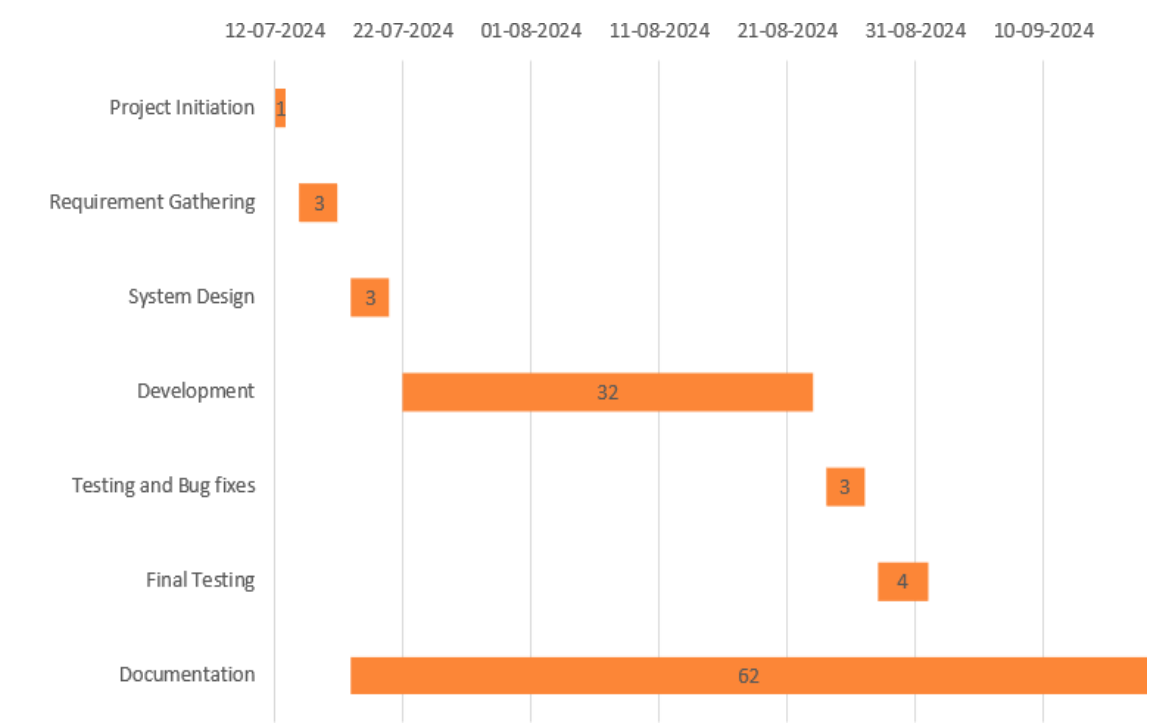
3.3 Analysis of the Proposed System

The current manual systems used in managing NCC units have several challenges that affect efficiency and accuracy. These systems depend on paperwork and human effort, which slows down processes and increases the chance of errors. Heavy workloads and poor communication make it difficult to manage large data and keep everyone informed in a timely manner. Understanding these problems will help us design a better, automated system for improving management and communication.

Some key problems are:

- Manual data entry leads to mistakes and missing records.
- Slow updates and delays hinder timely communication.
- Staff face heavy workloads, causing fatigue and inefficiency.
- Difficult to manage and organize large volumes of cadet and event data.
- Lack of real-time notifications causes communication gaps.
- Reporting is inaccurate and delayed, affecting planning and monitoring

3.4 Gantt Chart



Task	Start date	End Date	Duration(Days)
Project Initiation	12-07-2024	13-07-2024	1
Requirement Gathering	14-07-2024	17-07-2024	3
System Design	18-07-2024	21-07-2024	3
Development	22-07-2024	23-08-2024	32
Testing and Bug fixes	24-08-2024	27-08-2024	3
Final Testing	28-08-2024	01-09-2024	4
Documentation	18-07-2024	18-09-2024	62

CHAPTER 4. SURVEY OF TECHNOLOGY

4.1 HTML

Hypertext Markup Language, or HTML, is the fundamental language used to create the structure and content of web pages. It consists of tags enclosed in angle brackets

that tell the web browser how to display the various elements of a page such as headings, paragraphs, lists, images, links, and forms. HTML is responsible for organizing these components in a logical, semantic way that defines the meaning and hierarchy of content on the page.

However, HTML by itself controls only the structure and content of the page. To make web pages visually attractive, the Cascading Style Sheets (CSS) language is used alongside HTML. CSS defines the look and feel of the page by controlling colors, fonts, layouts, spacing, and other stylistic features. With CSS, developers can make the same HTML content appear in many different styles without changing the document's structure.

While HTML and CSS handle structure and style, JavaScript adds interactivity and dynamic behavior to web pages. It enables pages to respond to user actions, modify content on the fly, validate forms, create animations, and enhance the overall user experience. JavaScript runs inside the browser and interacts with the HTML elements, changing content or styles based on events such as clicks or keystrokes.



4.2 CSS

CSS, or Cascading Style Sheets, is a crucial technology in web development used to control the appearance and layout of web pages. It allows developers to separate the content of a webpage, defined by HTML, from its design, such as colors, fonts, spacing, and positioning. This separation makes it possible to maintain and update websites more efficiently because the style rules can be changed independently from the content.

CSS works by applying styles through a system of rules consisting of selectors and declarations. Selectors target specific HTML elements, and declarations specify how those elements should be displayed, including properties like color, font size, margin, and padding. Styles can be added in three ways: directly inside HTML elements using inline styles, inside a `<style>` block within an HTML document, or most commonly, through external CSS files linked to the HTML page. The external file method enables a single stylesheet to style many pages

simultaneously, promoting consistency and reducing code duplication.

r

Advanced CSS layout techniques, such as Flexbox and Grid, give designers precise control over how elements are placed and how they respond to different screen sizes and devices. This adaptability is vital for creating responsive designs that provide a seamless user experience on desktops, tablets, and smartphones.

Overall, CSS enhances the visual appeal and usability of websites, making it an indispensable tool in modern web design and development. By controlling aesthetics separately from structure, CSS empowers developers to create beautiful, flexible, and accessible web interfaces.



4.3 JavaScript

JavaScript is a versatile and widely-used programming language that plays a crucial role in modern web development. Unlike HTML and CSS, which focus on the structure and style of web pages, JavaScript adds interactivity and dynamic behavior to web content. It is a client-side scripting language, meaning that the code runs directly in the user's web browser without needing to communicate with the server for every action. This allows web pages to respond instantly to user actions such as clicks, form inputs, and other events.

JavaScript can manipulate the Document Object Model (DOM), which represents the structure of an HTML page, enabling developers to dynamically change content, styles, and attributes in real-time. For example, it can modify text, images, and layout or create complex features like sliders, pop-up windows, and real-time form validation. JavaScript also supports asynchronous programming, which is essential for handling network requests without freezing the interface. This capability powers dynamic content loading, live updates, and smooth user experiences.

Beyond its traditional use in browsers, JavaScript has expanded into server-side applications through platforms like Node.js, enabling full-stack development with a single language. It is also widely used in game development and mobile app development using frameworks such as React Native. JavaScript's vast ecosystem of libraries and frameworks enhances its power and flexibility, making it indispensable for creating interactive, scalable, and modern web applications.

JavaScript



4.4 MongoDB Database

MongoDB is a modern NoSQL database designed to handle large volumes of data with high flexibility and performance. Unlike traditional relational databases that use tables and fixed schemas, MongoDB stores data in JSON-like documents, allowing for a schema-less design. This flexibility lets developers easily add or change fields in a document without complex database migrations, making it ideal for projects where data evolves frequently.

Key features of MongoDB include horizontal scalability through sharding, which distributes data across multiple servers to ensure smooth performance as data grows. It also provides high availability with replication, creating copies of data on different servers to prevent data loss and maintain uptime. MongoDB supports advanced querying and indexing, enabling fast and efficient data retrieval even from large datasets.

The document-oriented data model allows MongoDB to natively store complex, nested data structures without costly join operations, simplifying development and improving speed. This architecture is particularly useful for applications that need to store varied user information, logs, or feedback, like your NCC Unit Management Website.

Overall, MongoDB offers a combination of flexibility, scalability, and performance, making it a strong choice for modern, data-intensive applications that need to adapt and grow quickly.



mongoDB

4.4 Python Django

Django is a high-level Python web framework designed to enable rapid and clean web development. It follows the Model-Template-View (MTV) architecture, which helps separate data handling, user interface, and business logic efficiently. Django comes with many built-in features such as an automatic admin panel, a powerful Object-Relational Mapping (ORM) system, form handling, and user authentication. These features reduce development time by providing ready-to-use components and tools out of the box.

Django is known for its strong security measures, protecting applications from common threats like SQL injection, cross-site scripting (XSS), and cross-site request forgery (CSRF). It also supports scalable and maintainable applications, making it suitable for small to large projects. The framework's extensive documentation and large community support further help developers solve problems quickly and build reliable applications.

For your NCC Unit Management Website, Django's built-in features like user authentication, database integration, and admin interface accelerate development while ensuring security and performance. Django's flexibility also allows easy customization to meet specific project requirements.



CHAPTER 5. SYSTEM DESIGN

5.1 Introduction

In today's fast-changing world of medical technology, designing an effective website system is vital. This step bridges the gap between idea and reality, showing how the website will work and what features it will have. A well-planned system design helps doctors and healthcare staff manage important activities comfortably, such as risk assessments, resource management, patient communication, and accurate diagnosis. The aim is to create an easy-to-use and dependable website that supports healthcare professionals while benefiting the patients they serve.

Our website design focuses on clear organization, smooth data flow, and efficient communication. It will store important information, like patient records, appointment schedules, treatment details, and feedback securely. The system will let doctors quickly access data, reducing errors and saving time. It will also make it easy to communicate between departments and with patients, improving overall care.

Good system design also means the website will be flexible and scalable. As patient numbers grow and new features become necessary, the website will adapt without losing speed or reliability. User-friendly pages and simple navigation will ensure that medical staff and users of all skill levels can use the site with ease. Security and privacy are prioritized to protect sensitive health information.

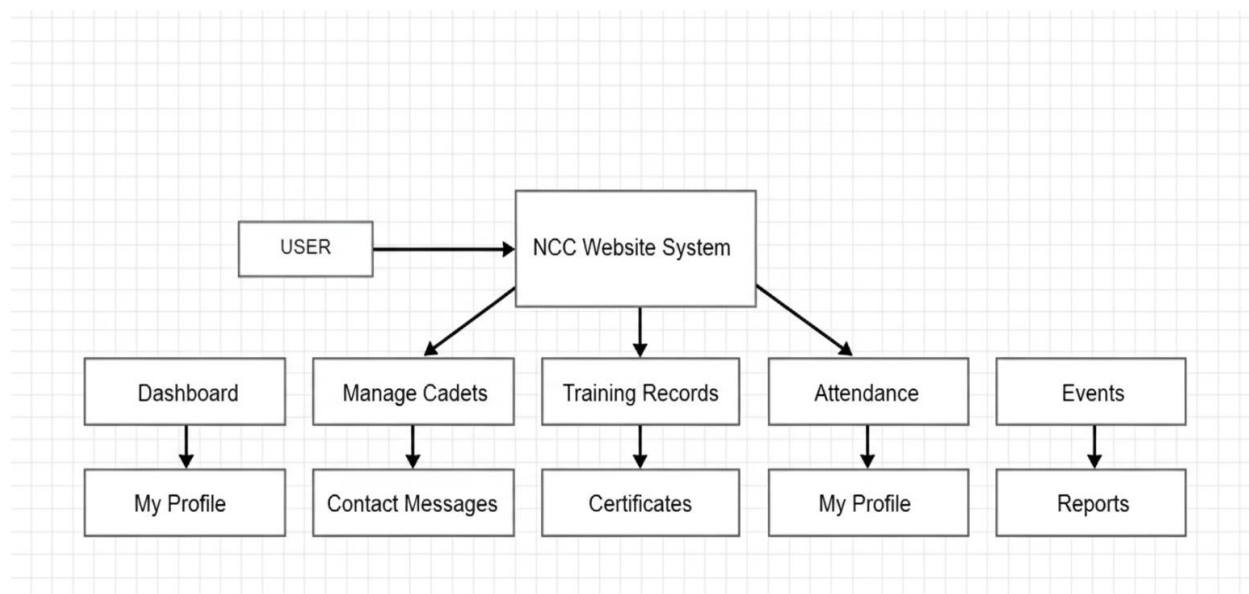
Overall, the system design sets the foundation for a powerful healthcare digital tool. It combines technology and user needs carefully to create a website that streamlines medical workflows, enhances patient care, and supports healthcare providers in their mission to serve the community effectively.

5.2 System Architecture Diagram

System architecture diagram is a visual tool that represents the structure of a system. It shows how different parts of the system connect and interact, and what role each part plays. By looking at this diagram, one can understand the main functions of the system and how these functions work together to achieve the overall goals.

In the context of an NCC Unit Management Website, the system architecture diagram will illustrate the flow of data and the interaction between components such as the user interface, authentication service, registration module, attendance tracking, event enrollment, feedback collection, the admin panel, the database backend, and notification services.

Each component performs specific functions that, when combined, create a seamless experience for the user. For example, the user logs in through the authentication service, accesses the dashboard to fill registration forms or submit attendance, enrolls in events, and submits feedback. Admins use a separate panel to manage this data, such as approving attendance or viewing feedback. The database stores all these details securely. Notifications keep users updated about important activities like event schedules or deadlines.

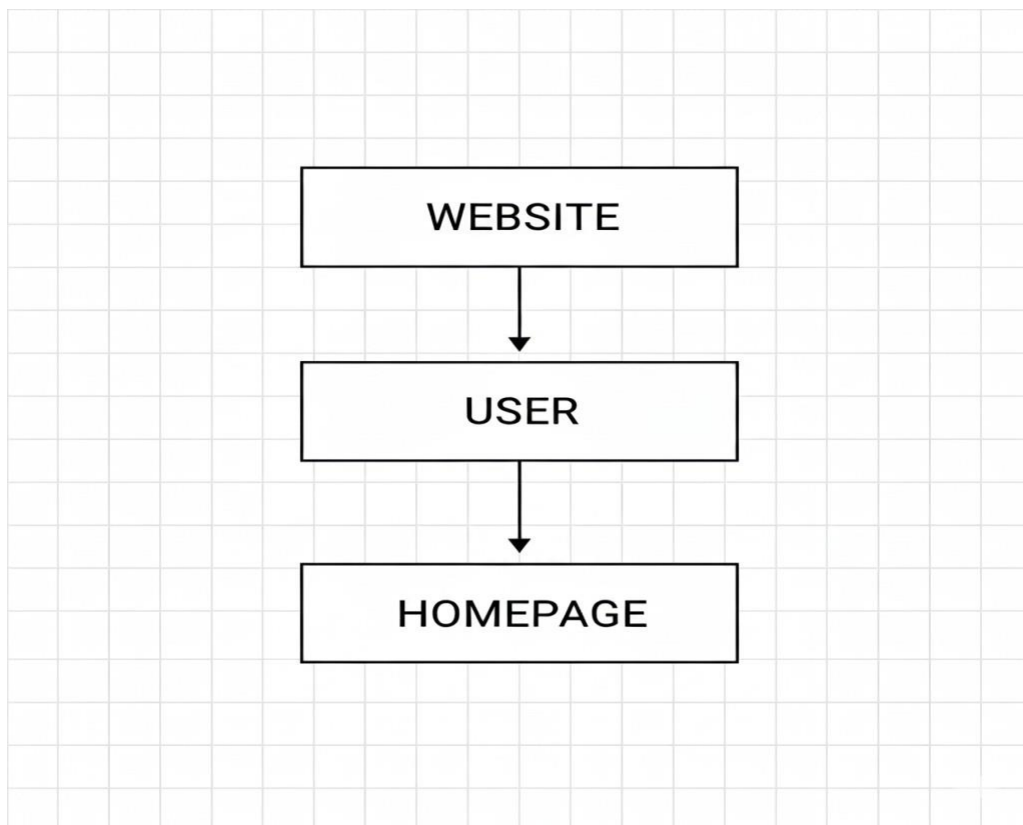


5.3 Data Flow Diagram

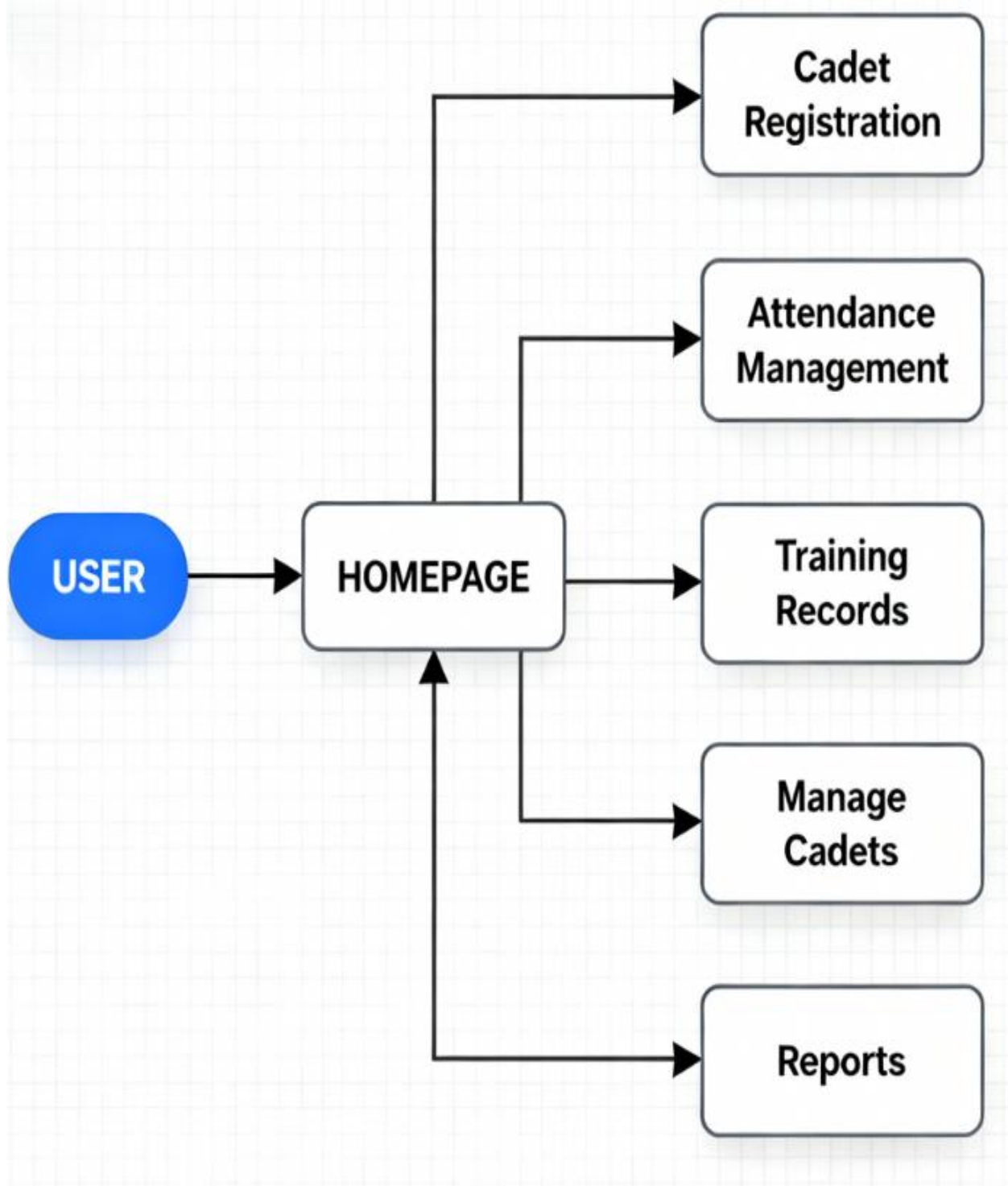
A Data Flow Diagram (DFD) is a graphical representation that shows how data moves through a system. It highlights the processes that handle data, the flow of data between these processes, the storage points, and where external entities interact with the system. Essentially, a DFD visualizes the flow of information from input to storage and finally to output or report generation.

DFDs help in understanding how information is processed and transferred within the system. For example, in the NCC Unit Management Website, data flows might include user registration details moving from the registration form to the database, attendance records being submitted and stored, or feedback being collected and later accessed by administrators for analysis.

Data Flow Diagram Level-0

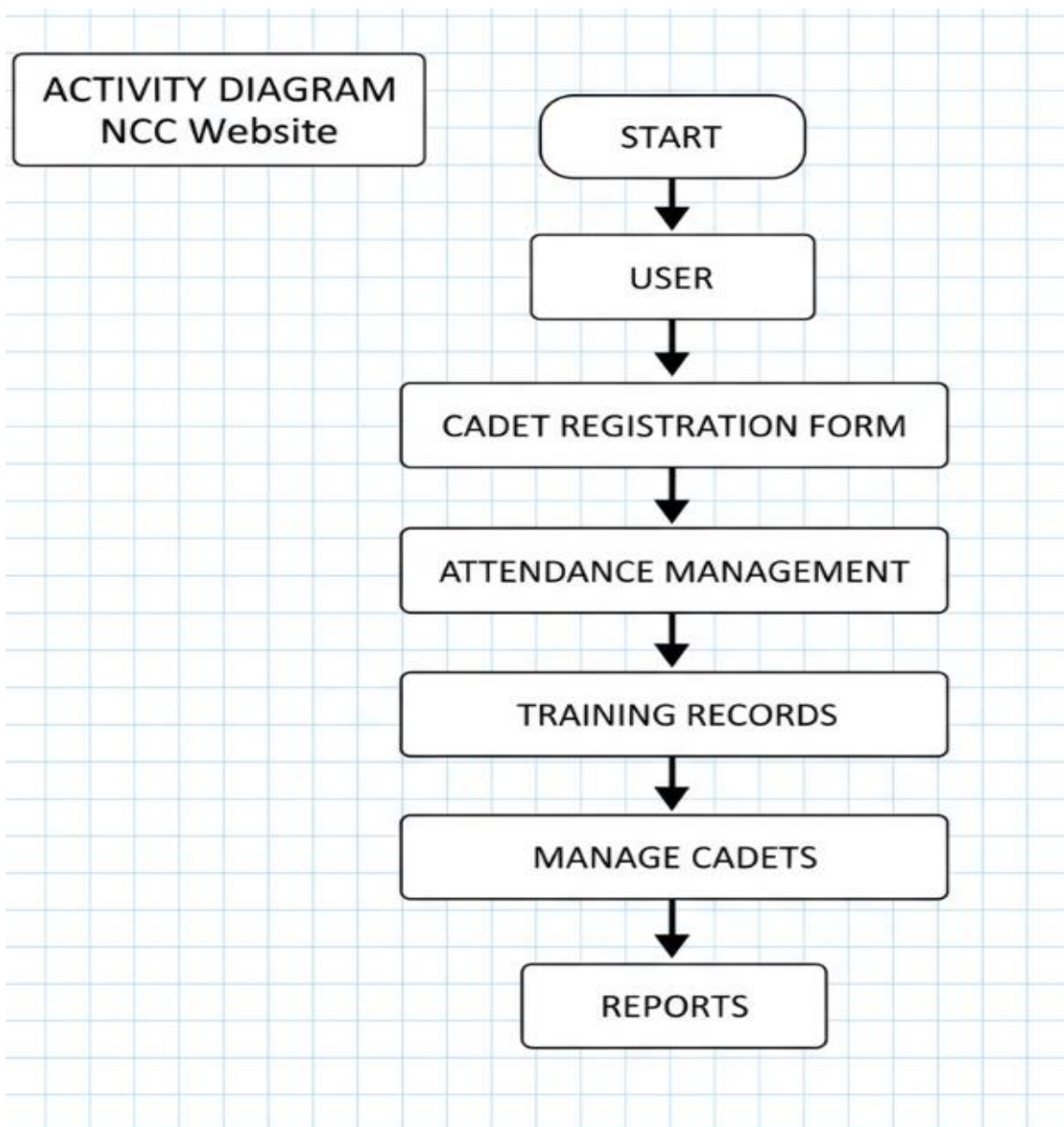


Data Flow Diagram Level-1 (User)



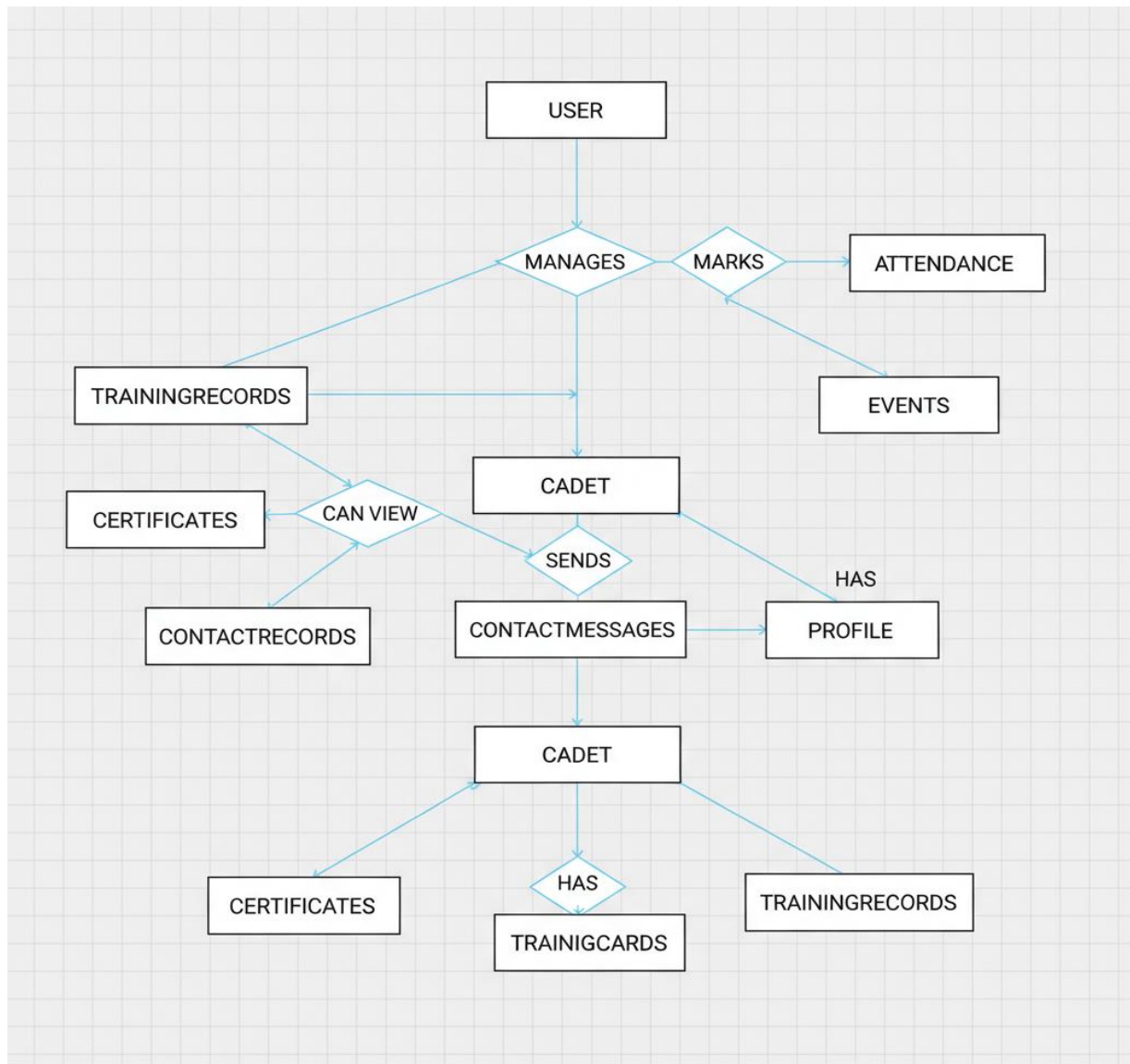
5.4Activity Diagram

An activity diagram is a type of UML (Unified Modeling Language) diagram used to visualize and model the flow of activities, actions, and decisions within a system, process, or workflow. It helps in depicting the sequential and parallel activities, along with decision points and control flows, making it a valuable tool for understanding, documenting, and improving processes or systems in various domains, such as software development, business processes, and project management.



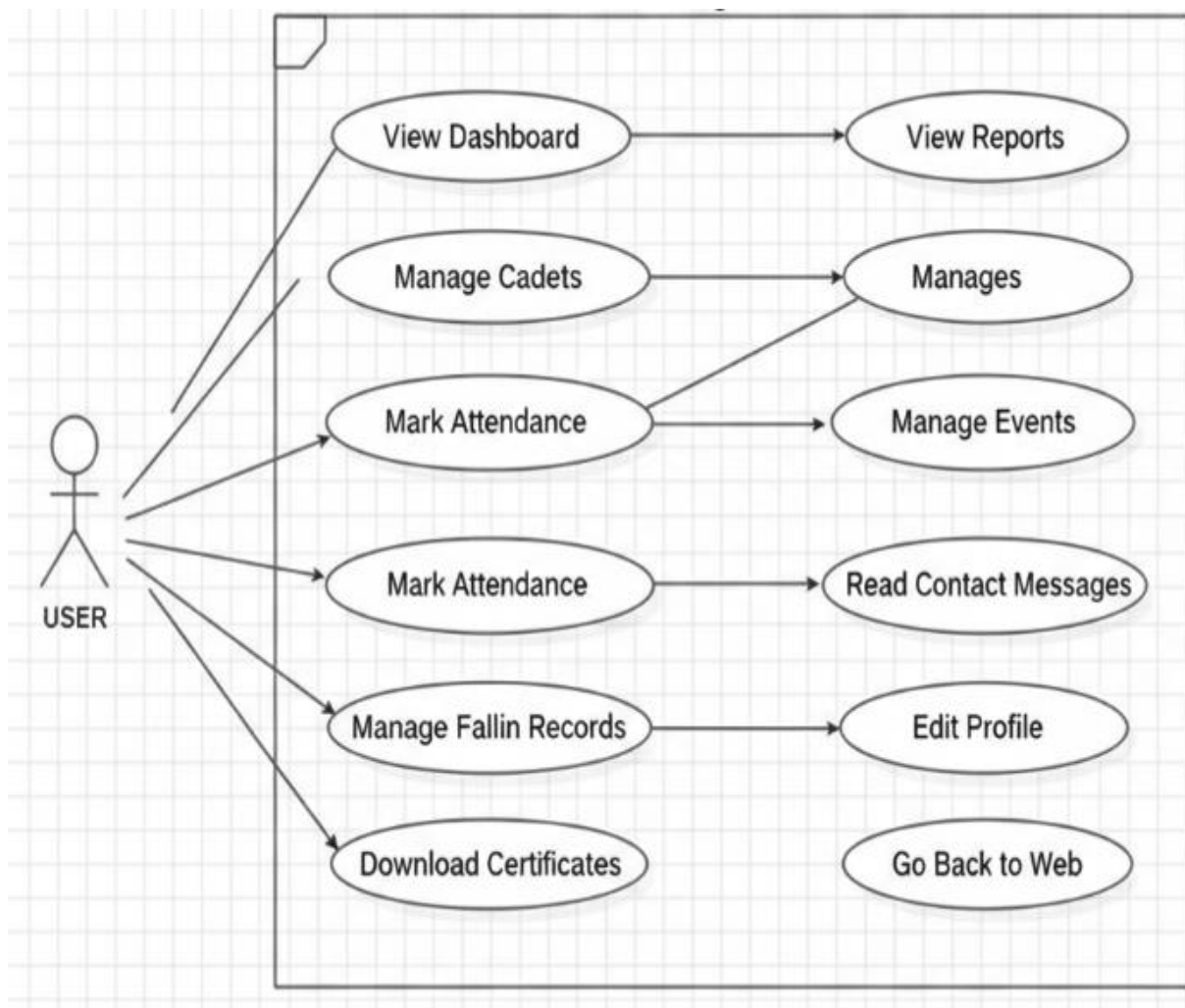
5.5E-R Diagram

An ER diagram shows the relationship among entity sets. An entity set is a group of similar entities and these entities can have attributes. In terms of DBMS, an entity is a table or attribute of a table in database, so by showing relationship among tables and their attributes, ER diagram shows the complete logical structure of a database. Entity Relational (ER) Model is a highlevel conceptual data model diagram. ER modelling helps you to analyse data requirements systematically to produce a well-designed database. The Entity-Relation model represents realworld entities and the relationship between them.



5.6 Use-Case Diagram

A use case diagram is a visual representation in Unified Modelling Language (UML) that illustrates the various ways an external actor interacts with a system or software application. It defines the functional requirements of the system by depicting different use cases (or actions) that the system must perform in response to user or external system inputs, helping to understand how the system will be used from a user's perspective and providing a high-level overview of its functionality.



CHAPTER 6. SYSTEM IMPLEMENTATION

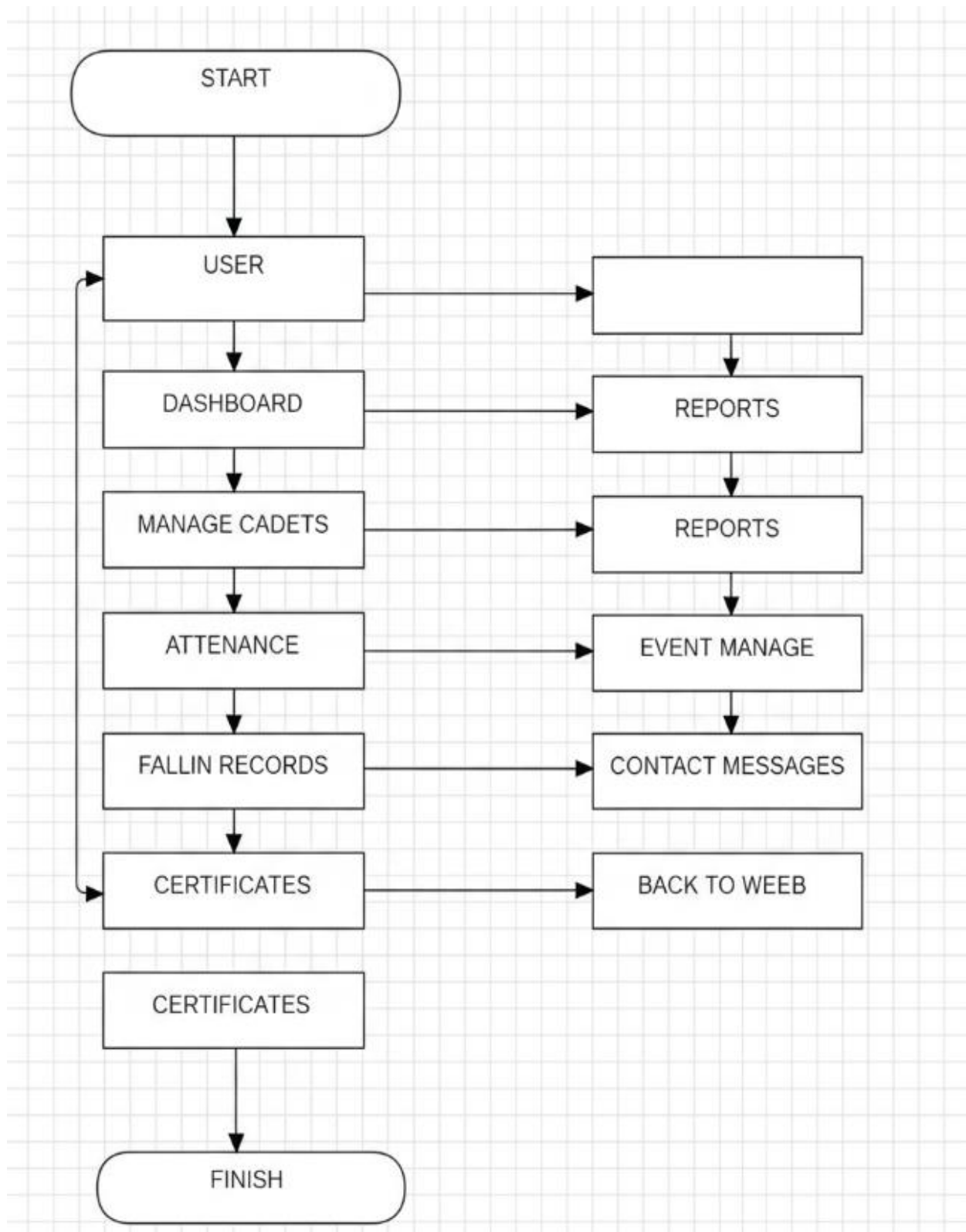
6.1 Introduction

System implementation in the context of the NCC Unit Management Website marks the transformation of all project planning, design, and coding into a fully functional digital platform for effective unit management. This phase involves deploying all backend and frontend components using technologies such as Django, Python, and MongoDB to ensure seamless operation.

During implementation, each module—like Cadet Registration, Attendance Management, Training Records, Reports, Events, and Profile handling—was incrementally integrated and rigorously tested in the development environment. Data flows between forms and the database were established, ensuring that features such as student/teacher registration, real-time attendance, and event management work reliably and efficiently.

The implementation process also included configuring security (like authentication and user roles), setting up the user interface in line with academic and user-friendly standards, and integrating feedback features. Extensive testing was conducted to validate all functionalities, eliminate bugs, and ensure the site met both user expectations and stated objectives. User documentation and support resources were additionally created to aid new users in navigating the platform.

6.2 Flowchart



6.3 Coding

Home.html

```
{% extends 'base.html' %}
{% block title %}Home - Rizvi NCC Unit{% endblock %}

{% block content %}
{% load static %}
<style>
/* Responsive styles for home page */
.hero {
  padding: 40px 0;
}
.hero-left {
  background-color: #2c3e5c;
  color: #f5b21a;
  border-radius: 12px;
  box-shadow: 0 4px 12px rgba(0,0,0,0.3);
  padding: 40px 20px;
  text-align: center;
  margin-bottom: 20px;
}
.hero-left .sub {
  font-size: 1rem;
  font-weight: 600;
  margin-bottom: 15px;
  letter-spacing: 1px;
}
.hero-left h1 {
  font-size: 2.5rem;
  font-weight: 900;
  line-height: 1.2;
  margin-bottom: 15px;
}
.hero-left .badge {
```

```
width: 120px;
height: 120px;
border: 2px solid #f5b21a;
border-radius: 50%;
display: inline-flex;
justify-content: center;
align-items: center;
font-weight: 700;
font-size: 1rem;
margin-top: 20px;
}
@media (min-width: 768px) {
  .hero-left {
    padding: 60px;
    text-align: left;
    margin-bottom: 0;
  }
  .hero-left h1 {
    font-size: 3.5rem;
  }
  .hero-left .badge {
    width: 140px;
    height: 140px;
    font-size: 1.2rem;
  }
}
.hero-right {
  background-color: #7db9e8;
  color: white;
  border-radius: 12px;
  box-shadow: 0 4px 12px rgba(0,0,0,0.3);
  padding: 40px 20px;
  text-align: center;
}
.hero-right h2 {
```

```
font-weight: 900;
font-size: 2.5rem;
margin-bottom: 20px;
}
.hero-right p {
font-weight: 400;
font-size: 1.2rem;
margin-bottom: 30px;
}
.hero-right .btn-container {
display: flex;
flex-wrap: wrap;
justify-content: center;
gap: 15px;
}
.hero-right a {
padding: 15px 30px;
font-weight: 700;
text-decoration: none;
border-radius: 4px;
display: inline-block;
}
@media (min-width: 768px) {
.hero-right {
padding: 60px;
text-align: left;
}
.hero-right h2 {
font-size: 3rem;
margin-bottom: 40px;
}
.hero-right p {
font-size: 1.6rem;
margin-bottom: 60px;
}
```

```
.hero-right .btn-container {
  justify-content: flex-start;
}
.hero-right a {
  padding: 25px 50px;
  margin-right: 25px;
}
}
.btn-black {
  background-color: black;
  color: white;
}
.btn-outline {
  background-color: white;
  color: black;
}
/* Faculty Profile Section */
.faculty-profile {
  background-color: #2c3e5c;
  padding: 40px 20px;
  border-radius: 12px;
  box-shadow: 0 10px 30px rgba(0,0,0,0.25);
  color: #fff;
  margin: 40px 0;
}
.faculty-profile img {
  width: 100%;
  height: auto;
  border-radius: 12px;
  border: 2px solid #f5b21a;
  max-width: 300px;
  margin-bottom: 20px;
}
.faculty-details h3 {
  margin-bottom: 20px;
```

```
    text-align: center;
}
.faculty-details p {
    margin: 6px 0;
}
@media (min-width: 768px) {
    .faculty-profile img {
        margin-bottom: 0;
    }
    .faculty-details h3 {
        text-align: left;
    }
}
/* Footer Contact Info Section */
.footer-contact {
    background-color: #7db9e8;
    padding: 20px;
    text-align: center;
    border-top: 2px solid #f5b21a;
    margin-top: 40px;
}
.footer-contact button {
    background-color: #f5b21a;
    color: #2c3e5c;
    border: none;
    padding: 10px 20px;
    font-weight: 700;
    cursor: pointer;
}
</style>
```

```
<!-- Hero Section -->
<section class="hero">
  <div class="container">
    <div class="row">
```

```

<div class="col-12 col-md-6 hero-left">
  <div class="sub">RIZVI EDUCATION SOCIETY'S</div>
  <h1>RIZVI COLLEGE<br>OF ARTS, SCIENCE<br>AND COMMERCE</h1>
  <div class="sub">NATIONAL CADET CORPS (NCC)<br>1 MAH BATTALION NCC,
MUMBAI-B ALPHA COY</div>
  <div class="badge" style="border-radius: 12px; width: 180px; height: 180px; display: flex;
justify-content: center; align-items: center; font-weight: 700; font-size: 1rem; margin: 20px auto
0 auto;">
    
  </div>
</div>
<div class="col-12 col-md-6 hero-right">
  <h2>START YOUR DEFENCE JOURNEY<br>WITH RIZVI NCC UNIT</h2>
  <p>Endurance is the ability to withstand hard work, stress, and hardship. It is the key to
becoming a true warrior.</p>
  <div class="btn-container" style="gap: 0;">
    <a href="{% url 'register' %}" class="btn-black" style="flex: 1; text-align:
center;">REGISTER AS CADET</a>
    <a href="{% url 'login' %}" class="btn-outline" style="flex: 1; text-align: center; font-
weight: 900; font-size: 1.2rem;">LOGIN</a>
  </div>
</div>
</div>
</section>

```

<!-- Faculty Profile Section -->

<section class="faculty-profile">

<div class="container">

<div class="row align-items-center">

<div class="col-12 col-md-4 text-center">



Faculty Details

Name	Department	Contact
Dr. John Doe	Computer Science	(907) 555-1234
Prof. Jane Smith	Biology	(907) 555-2345
Dr. Michael Chen	Physics	(907) 555-3456
Dr. Emily White	Chemistry	(907) 555-4567


```
<h3>Associate NCC Officer (ANO)</h3>
<p><strong>Name:</strong> Roshan Khobragade </p>
<p><strong>Qualifications:</strong> BA.,LLB.,LLM.,NET.,Mphil </p>
<p><strong>Department:</strong> Business Law & National Cadet Corps (NCC) </p>
<p><strong>Designation:</strong> Associate NCC Officer (ANO) & Assistant Professor
</p>
<p><strong>Teaching Experience:</strong> 12 years & 9 Year As (ANO)</p>
<p><strong>Specializations:</strong> Law , NCC Training, Leadership Development </p>
</div>
</div>
</div>
</section>
```

```
<!-- Footer Contact Info Section -->
```

```
<section class="footer-contact">
  <div class="container">
    <button id="contactUsBtn" type="button" class="btn btn-warning">Contact Us</button>
  </div>
</section>
```

```
<!-- Contact Form Modal -->
```

```
<div class="modal fade" id="contactModal" tabindex="-1" aria-
labelledby="contactModalLabel" aria-hidden="true">
  <div class="modal-dialog modal-dialog-centered">
    <div class="modal-content text-dark">
      <div class="modal-header">
        <h5 class="modal-title" id="contactModalLabel">Contact Us</h5>
        <button type="button" class="btn-close" data-bs-dismiss="modal" aria-
label="Close"></button>
      </div>
      <div class="modal-body">
        <form id="contactForm" method="post" novalidate>
          {% csrf_token %}
          <div class="mb-3">
            <input type="text" name="name" class="form-control" placeholder="Your Name"
```

```

required>
    <div class="invalid-feedback">Please enter your name.</div>
</div>
<div class="mb-3">
    <input type="email" name="email" class="form-control" placeholder="Your Email"
required>
    <div class="invalid-feedback">Please enter a valid email.</div>
</div>
<div class="mb-3">
    <input type="text" name="subject" class="form-control" placeholder="Subject"
required>
    <div class="invalid-feedback">Please enter a subject.</div>
</div>
<div class="mb-3">
    <textarea name="message" rows="5" class="form-control" placeholder="Your Message"
required></textarea>
    <div class="invalid-feedback">Please enter your message.</div>
</div>
<div id="formAlert" class="alert d-none" role="alert"></div>
<button type="submit" class="btn btn-warning w-100">Send Message</button>
</form>
</div>
</div>
</div>
</div>

<script>
(function() {
    const form = document.getElementById('contactForm');
    const modalEl = document.getElementById('contactModal');
    const formAlert = document.getElementById('formAlert');
    const contactUsBtn = document.getElementById('contactUsBtn');
    const modal = new bootstrap.Modal(modalEl);

    contactUsBtn.addEventListener('click', () => {

```

```

    modal.show();
  });

form.addEventListener('submit', function(event) {
  event.preventDefault();
  formAlert.classList.add('d-none');
  formAlert.textContent = "";
  formAlert.classList.remove('alert-success', 'alert-danger');

  // Simple client-side validation
  let valid = true;
  Array.from(form.elements).forEach(el => {
    if (el.hasAttribute('required') && !el.value.trim()) {
      el.classList.add('is-invalid');
      valid = false;
    } else {
      el.classList.remove('is-invalid');
    }
  });
  if (!valid) return;

  const formData = new FormData(form);

  fetch(`${% url 'contact' %}`, {
    method: 'POST',
    headers: {
      'X-Requested-With': 'XMLHttpRequest',
      'X-CSRFToken': formData.get('csrfmiddlewaretoken'),
    },
    body: formData
  })
  .then(response => {
    if (!response.ok) throw new Error('Network response was not ok');
    return response.json();
  })

```

```
.then(data => {
  if (data.success) {
    formAlert.textContent = data.message || 'Message sent successfully!';
    formAlert.classList.remove('d-none');
    formAlert.classList.add('alert-success');
    form.reset();
    setTimeout(() => {
      modal.hide();
      formAlert.classList.add('d-none');
    }, 2000);
  } else {
    formAlert.textContent = data.error || 'Failed to send message.';
    formAlert.classList.remove('d-none');
    formAlert.classList.add('alert-danger');
  }
})
.catch(error => {
  formAlert.textContent = 'An error occurred. Please try again later.';
  formAlert.classList.remove('d-none');
  formAlert.classList.add('alert-danger');
});
})();
</script>
{% endblock %}
```

Dashboard.html

```
{% extends 'base_dashboard.html' %}
{% load static %}
{% block title %}Dashboard - NCC Management{% endblock %}
```

```
{% block content %}
```

```
<style>
```

```
/* Dashboard overview */
```

```
.dashboard-overview {
    background-color: #fefefe;
    box-shadow: 0 2px 8px rgb(0 0 0 / 0.1);
    border-radius: 8px;
    padding: 20px;
    margin-bottom: 30px;
}
```

```
.dashboard-overview h2 {
    font-weight: 700;
    margin-bottom: 20px;
    font-size: 1.5rem;
    color: #34495e;
}
```

```
.overview-cards {
    display: flex;
    gap: 20px;
    flex-wrap: wrap;
}
```

```
.overview-card {
    flex: 1 1 200px;
    background-color: white;
    border-radius: 8px;
    padding: 20px;
    box-shadow: 0 0 3px transparent;
    transition: box-shadow 0.3s ease;
```

```
    text-align: center;
}
.overview-card:nth-child(odd) {
    box-shadow: 0 0 0 3px #3498db;
}
.overview-card h3 {
    font-weight: 700;
    font-size: 2rem;
    color: #2980b9;
    margin-bottom: 8px;
}
.overview-card p {
    font-weight: 600;
    color: #7f8c8d;
}
/* Quick actions */
.quick-actions {
    margin-bottom: 30px;
}
.quick-actions h2 {
    font-weight: 700;
    margin-bottom: 20px;
    font-size: 1.3rem;
    color: #34495e;
}
.actions-container {
    display: flex;
    gap: 20px;
    flex-wrap: wrap;
}
.action-card {
    flex: 1 1 300px;
    background-color: white;
    border-radius: 8px;
    padding: 30px 20px;
```

```
    box-shadow: 0 2px 8px rgb(0 0 0 / 0.1);
    text-align: center;
}
.action-icon {
    font-size: 40px;
    margin-bottom: 15px;
    color: #5d6d7e;
}
.action-title {
    font-weight: 700;
    font-size: 1.1rem;
    margin-bottom: 8px;
    color: #2c3e50;
}
.action-desc {
    font-size: 0.9rem;
    color: #7f8c8d;
    margin-bottom: 15px;
}
.action-btn {
    padding: 10px 20px;
    border: none;
    border-radius: 5px;
    font-weight: 600;
    cursor: pointer;
    color: white;
    transition: background-color 0.3s ease;
    text-decoration: none;
    display: inline-block;
}
.btn-manage {
    background-color: #34495e;
}
.btn-manage:hover {
    background-color: #2c3e50;
```

```
}  
.btn-attendance {  
  background-color: #27ae60;  
}  
.btn-attendance:hover {  
  background-color: #229954;  
}  
.btn-training {  
  background-color: #f39c12;  
}  
.btn-training:hover {  
  background-color: #d68910;  
}  
.btn-events {  
  background-color: #9b59b6;  
}  
.btn-events:hover {  
  background-color: #8e44ad;  
}  
  
/* Responsive styles for small screens */  
@media (max-width: 768px) {  
  .overview-cards {  
    flex-direction: column;  
    gap: 15px;  
  }  
  .overview-card {  
    flex: 1 1 100%;  
    padding: 15px;  
  }  
  .actions-container {  
    flex-direction: column;  
    gap: 15px;  
  }  
  .action-card {
```



```
    flex: 1 1 100%;
    padding: 20px 15px;
}
.action-icon {
    font-size: 30px;
    margin-bottom: 10px;
}
.action-title {
    font-size: 1rem;
}
.action-desc {
    font-size: 0.85rem;
}
.action-btn {
    padding: 8px 16px;
    font-size: 0.9rem;
}
}
```

</style>

```
{% if user_type == 'ano' %}
<section class="dashboard-overview">
  <h2>Dashboard Overview</h2>
  <div class="overview-cards">
    <div class="overview-card">
      <h3>{{ total_cadets }}</h3>
      <p>Total Cadets</p>
    </div>
    <div class="overview-card">
      <h3>{{ active_cadets }}</h3>
      <p>Active Cadets</p>
    </div>
    <div class="overview-card">
      <h3>{{ training_programs_count }}</h3>
```

```

        <p>Training Programs</p>
    </div>
    <div class="overview-card">
        <h3>{{ upcoming_events_count }}</h3>
        <p>Upcoming Events</p>
    </div>
</div>
</section>
{% endif %}

<section class="quick-actions">
    <h2>Quick Actions</h2>
    <div class="actions-container">
        <div class="action-card">
            <div class="action-icon"><img alt="User icon" data-bbox="361 401 381 421"/></div>
            <div class="action-title">My Profile</div>
            <div class="action-desc">View and manage your personal profile</div>
            <a href="{% url 'profile' %}" class="action-btn btn-manage">View Profile</a>
        </div>
        <div class="action-card">
            <div class="action-icon"><img alt="Group of people icon" data-bbox="361 554 381 574"/></div>
            <div class="action-title">Cadets Profile</div>
            <div class="action-desc">view all cadets profile of RIZVI NCC UNIT</div>
            <a href="{% url 'cadets' %}" class="action-btn btn-manage">Manage Cadets</a>
        </div>
        {% if user_type == 'ano' %}
        <div class="action-card">
            <div class="action-icon"><img alt="Clipboard icon" data-bbox="361 728 381 748"/></div>
            <div class="action-title">Take Attendance</div>
            <div class="action-desc">Mark today's attendance for training sessions</div>
            <a href="{% url 'attendance' %}" class="action-btn btn-attendance">Take Attendance</a>
        </div>
        {% endif %}
        <div class="action-card">
            <div class="action-icon"><img alt="Globe icon" data-bbox="361 902 381 922"/></div>

```

```

<div class="action-title">FALLIN SCHEDULE</div>
<div class="action-desc">view fallin schedule</div>
  <a href="{% url 'training' %}" class="action-btn btn-training">View Training</a>
</div>
<div class="action-card">
  <div class="action-icon">

```

```

        <th style="padding:8px;">Username</th>
        <th style="padding:8px;">Name</th>
        <th style="padding:8px;">Reg No</th>
        <th style="padding:8px;">Phone</th>
        <th style="padding:8px;">Actions</th>
    </tr>
</thead>
<tbody>
    {% for cadet in cadets %}
    <tr style="border-bottom:1px solid #f1f1f1;">
        <td style="padding:8px;">{{ cadet.user.username }}</td>
        <td style="padding:8px;">{{ cadet.name }}</td>
        <td style="padding:8px;">{{ cadet.reg_no }}</td>
        <td style="padding:8px;">{{ cadet.phone }}</td>
        <td style="padding:8px;">
            <a href="{% url 'edit_cadet' user_id=cadet.user.id %}" class="btn btn-sm btn-warning"
style="margin-right:8px;">Edit</a>
            <form method="post" action="{% url 'delete_cadet' user_id=cadet.user.id %}"
style="display:inline-block;" onsubmit="return confirm('Are you sure you want to delete this
cadet?');">
                {% csrf_token %}
                <button type="submit" class="btn btn-sm btn-danger">Delete</button>
            </form>
        </td>
    </tr>
    {% endfor %}
</tbody>
</table>
{% else %}
    <p>No cadets registered yet.</p>
{% endif %}
</div>
</section>
{% endif %}
{% endblock %}

```

Dashboard_views.py

```
from django.contrib.auth.decorators import login_required
from django.http import HttpResponseRedirect, HttpResponseForbidden,
HttpResponseNotAllowed
from django.http import JsonResponse
from django.shortcuts import redirect, get_object_or_404, render
from django.template.loader import render_to_string
from django.middleware.csrf import get_token
from .models import UserProfile, Attendance, Event, TrainingProgram, Cadet
from django.contrib.auth.models import User
from datetime import date

@login_required
def dashboard_view(request):
    from .models import Cadet
    try:
        user_profile = UserProfile.objects.get(user=request.user)
    except UserProfile.DoesNotExist:
        user_profile = None

    if request.method == 'POST' and user_profile and user_profile.user_type == 'ano':
        cadets = Cadet.objects.all()
        for cadet in cadets:
            status = request.POST.get(f'attendance_{cadet.user.id}')
            if status in ['present', 'absent', 'camp']:
                Attendance.objects.update_or_create(
                    cadet=cadet,
                    date=date.today(),
                    defaults={'status': status}
                )
        return redirect('dashboard')

    total_cadets = Cadet.objects.count()
    active_cadets = Cadet.objects.filter(user__userprofile__user_type='cadet').count()
    training_programs_count = TrainingProgram.objects.filter(is_active=True).count()
    upcoming_events_count = Event.objects.filter(is_upcoming=True,
date__gte=date.today()).count()
    cadets = Cadet.objects.all()

    # Adjust context based on user type
    context = {
        'user': request.user,
        'user_profile': user_profile,
        'username': request.user.username,
```

```

        'user_type': user_profile.user_type if user_profile else "",
        'total_cadets': total_cadets,
        'active_cadets': active_cadets,
        'training_programs_count': training_programs_count,
        'upcoming_events_count': upcoming_events_count,
        'cadets': cadets,
    }

# For non-ano users, remove attendance management visibility
if user_profile and user_profile.user_type != 'ano':
    context.pop('cadets', None)
    context.pop('total_cadets', None)
    context.pop('active_cadets', None)
    context.pop('training_programs_count', None)
    context.pop('upcoming_events_count', None)

return render(request, 'dashboard.html', context)

@login_required
def add_cadet_view(request: HttpRequest) -> HttpResponse:
    from .models import Cadet
    try:
        user_profile = UserProfile.objects.get(user=request.user)
    except UserProfile.DoesNotExist:
        return HttpResponseForbidden("User profile not found.")

    if user_profile.user_type != 'ano':
        return HttpResponseForbidden("Access denied. Only ANO can add cadets.")

    token = get_token(request)
    if request.method == 'POST':
        full_name = request.POST.get('full_name')
        email = request.POST.get('email')
        reg_no = request.POST.get('reg_no')
        rank = request.POST.get('rank')
        phone = request.POST.get('phone')
        date_joined = request.POST.get('date_joined')
        blood_group = request.POST.get('blood_group')
        is_active = request.POST.get('is_active') == 'on'
        image = request.FILES.get('image')

        if Cadet.objects.filter(reg_no=reg_no).exists():
            # Handle duplicate
            if request.headers.get('x-requested-with') == 'XMLHttpRequest':
                return JsonResponse({'success': False, 'errors': {'reg_no': ['Registration number already exists.']}}, status=400)

```

```

        # Non-AJAX: fallback to redirect (or you could show a message)
    else:
        # Create user with email as username and a default password
        user = User.objects.create_user(username=email, email=email,
password='defaultpassword')
        user.first_name = full_name
        user.save()
        UserProfile.objects.create(
            user=user,
            user_type='cadet',
            phone=phone
        )
        cadet = Cadet.objects.create(
            user=user,
            name=full_name,
            reg_no=reg_no,
            rank=rank,
            phone=phone,
            email=email,
            date_joined=date_joined,
            blood_group=blood_group,
            is_active=is_active,
            image=image
        )
        if request.headers.get('x-requested-with') == 'XMLHttpRequest':
            # Render partial cadet card and return HTML
            html = render_to_string('partials/cadet_card.html', {'cadet': cadet, 'user_type':
user_profile.user_type}, request=request)
            return JsonResponse({'success': True, 'html': html})
            return redirect('cadets')

    html = f'<!DOCTYPE html>
<html lang="en"><head><meta charset="UTF-8"><title>Add Cadet - NCC</title>
<style>
    * {{ margin: 0; padding: 0; box-sizing: border-box; }}
    body {{ font-family: 'Segoe UI', Arial, sans-serif; background: #f5f5f5; }}
    .top-navbar {{ background: linear-gradient(135deg, #2c3e50 0%, #34495e 100%); color:
white; padding: 15px 30px; display: flex; justify-content: space-between; align-items: center; }}
    .main-container {{ display: flex; min-height: calc(100vh - 70px); }}
    .sidebar {{ width: 250px; background: white; box-shadow: 2px 0 10px rgba(0,0,0,0.1);
padding: 20px 0; }}
    .sidebar-item {{ display: block; padding: 15px 25px; color: #555; text-decoration: none;
transition: all 0.3s; }}
    .sidebar-item:hover {{ background: #f8f9fa; color: #2c3e50; }}
    .content {{ flex: 1; padding: 30px; }}
    .form-container {{ background: white; padding: 30px; border-radius: 10px; box-shadow: 0

```

```

2px 15px rgba(0,0,0,0.08); max-width: 600px; }}
.form-group {{ margin-bottom: 20px; }}
label {{ display: block; margin-bottom: 8px; font-weight: 600; color: #333; }}
input, select {{ width: 100%; padding: 12px; border: 1px solid #ddd; border-radius: 5px; font-size: 16px; }}
.btn {{ padding: 12px 25px; background: #3498db; color: white; text-decoration: none; border-radius: 5px; font-size: 16px; border: none; cursor: pointer; }}
.btn:hover {{ background: #2980b9; }}
.btn-success {{ background: #27ae60; }}
.btn-success:hover {{ background: #219a52; }}
</style></head>
<body>
<nav class=\"top-navbar\"><div>🔔 NCC Management - Add Cadet</div><div><a href=\"/dashboard/\" style=\"color:white;margin-right:20px;\">← Back to Dashboard</a><a href=\"/logout/\" style=\"color:white;\">Logout</a></div></nav>
<div class=\"main-container\">
  <div class=\"sidebar\">
    <a href=\"/dashboard/\" class=\"sidebar-item\">🏠 Dashboard</a>
    <a href=\"/cadets/\" class=\"sidebar-item\">👤 Manage Cadets</a>
    <a href=\"/attendance/\" class=\"sidebar-item\">📅 Attendance</a>
    <a href=\"/training/\" class=\"sidebar-item\">🎯 Training Records</a>
    <a href=\"/reports/\" class=\"sidebar-item\">📊 Reports</a>
  </div>
  <div class=\"content\">
    <h1>Add New Cadet</h1>
    <div class=\"form-container\">
      <form method=\"post\" enctype=\"multipart/form-data\">
        <input type=\"hidden\" name=\"csrfmiddlewaretoken\" value=\"{token}\">
        <div class=\"form-group\"><label>Full Name:</label><input type=\"text\" name=\"full_name\" required></div>
        <div class=\"form-group\"><label>Email:</label><input type=\"email\" name=\"email\" required></div>
        <div class=\"form-group\"><label>Registration No:</label><input type=\"text\" name=\"reg_no\" required></div>
        <div class=\"form-group\"><label>Rank:</label><select name=\"rank\" required>
          <option value=\"Cadet\">Cadet</option>
          <option value=\"Lance Corporal\">Lance Corporal</option>
          <option value=\"Corporal\">Corporal</option>
          <option value=\"Sergeant\">Sergeant</option>
          <option value=\"Under Officer\">Under Officer</option>
        </select></div>
        <div class=\"form-group\"><label>Blood Group:</label><input type=\"text\" name=\"blood_group\" required></div>
        <div class=\"form-group\"><label>Phone:</label><input type=\"text\" name=\"phone\" required></div>
      </form>
    </div>
  </div>
</div>

```



```

        <div class="form-group"><label>Date Joined:</label><input type="date"
name="date_joined" required></div>
        <div class="form-group"><label>Photo:</label><input type="file" name="image"
accept="image/*"></div>
        <div class="form-group"><label><input type="checkbox" name="is_active"
checked> Active</label></div>
        <button type="submit" class="btn btn-success">Add Cadet</button>
    </form>
</div>
</div>
</div>
</body></html>""
    return HttpResponse(html)

```

@login_required

```
def delete_cadet_view(request: HttpRequest, user_id: int) -> HttpResponse:
```

```
    if request.method != 'POST':
```

```
        return HttpResponseNotAllowed(['POST'])
```

```
    try:
```

```
        user_profile = UserProfile.objects.get(user=request.user)
```

```
    except UserProfile.DoesNotExist:
```

```
        return HttpResponseForbidden("User profile not found.")
```

```
    if user_profile.user_type != 'ano':
```

```
        return HttpResponseForbidden("Access denied. Only ANO can delete cadets.")
```

```
    user_to_delete = get_object_or_404(User, id=user_id)
```

```
    try:
```

```
        cadet_to_delete = Cadet.objects.get(user=user_to_delete)
```

```
    except Cadet.DoesNotExist:
```

```
        return HttpResponseForbidden("Cadet profile not found.")
```

```
    # Delete the cadet and associated user
```

```
    cadet_to_delete.delete()
```

```
    user_to_delete.delete()
```

```
    return redirect('cadets')
```

@login_required

```
def cadets_view(request: HttpRequest) -> HttpResponse:
```

```
    from .models import Cadet
```

```
    try:
```

```
        user_profile = UserProfile.objects.get(user=request.user)
```

```
    except UserProfile.DoesNotExist:
```

```
        return HttpResponseForbidden("User profile not found.")
```

```

# Allow both ANO and Cadet users to view the list of cadets.
# Only ANO users will see edit/delete controls in the template.
if user_profile.user_type in ['ano', 'cadet']:
    cadets = Cadet.objects.all().order_by('name')
    total_cadets = cadets.count()
    return render(request, 'cadets.html', {'cadets': cadets, 'user_type': user_profile.user_type,
'total_cadets': total_cadets})
else:
    return HttpResponseForbidden("Access denied.")

from django.views.decorators.http import require_http_methods
from django.contrib import messages
from django.shortcuts import redirect

from collections import defaultdict
from datetime import datetime
from django.utils.timezone import localtime

@login_required
def training_view(request):
    from .models import Schedule
    try:
        user_profile = UserProfile.objects.get(user=request.user)
    except UserProfile.DoesNotExist:
        user_profile = None

    schedules = Schedule.objects.all().order_by('date', 'time')

    # Group schedules by date
    grouped_schedules = defaultdict(list)
    for schedule in schedules:
        grouped_schedules[schedule.date].append(schedule)

    # Sort dates
    sorted_dates = sorted(grouped_schedules.keys())

    # Prepare data for template: list of dicts with date and schedules
    training_data = []
    for date_key in sorted_dates:
        # Format date and day name
        date_str = date_key.strftime("%m/%d/%Y")
        day_name = date_key.strftime("%A")
        training_data.append({
            'date': date_str,
            'day_name': day_name,
            'schedules': grouped_schedules[date_key]

```

```

    })

    context = {
        'training_data': training_data,
        'user_type': user_profile.user_type if user_profile else "",
    }
    return render(request, 'training.html', context)

@login_required
@require_http_methods(["POST"])
def add_schedule_view(request):
    try:
        user_profile = UserProfile.objects.get(user=request.user)
    except UserProfile.DoesNotExist:
        return HttpResponseForbidden("User profile not found.")

    if user_profile.user_type != 'ano':
        return HttpResponseForbidden("Access denied. Only ANO can add schedules.")

    # Do NOT delete existing schedules before adding new ones (per user request)
    from .models import Schedule

    # Expecting multiple schedules in POST data as lists
    dates = request.POST.getlist('date')
    times = request.POST.getlist('time')
    activities = request.POST.getlist('activity')
    locations = request.POST.getlist('location')

    for date_str, time_str, activity, location in zip(dates, times, activities, locations):
        if date_str and time_str and activity and location:
            Schedule.objects.create(
                date=date_str,
                time=time_str,
                activity=activity,
                location=location
            )
    messages.success(request, "Fallin schedule updated successfully.")
    return redirect('training')

@login_required
@require_http_methods(["POST"])
def delete_schedule_view(request, schedule_id):
    try:
        user_profile = UserProfile.objects.get(user=request.user)
    except UserProfile.DoesNotExist:
        return HttpResponseForbidden("User profile not found.")

```

```
if user_profile.user_type != 'ano':
    return HttpResponseRedirect("Access denied. Only ANO can delete schedules.")
```

```
from .models import Schedule
schedule = get_object_or_404(Schedule, id=schedule_id)
schedule.delete()
messages.success(request, "Schedule entry deleted successfully.")
return redirect('training')
```

```
@login_required
def certificates_view(request):
    return render(request, 'certificates.html')
```

```
@login_required
def reports_view(request):
    return render(request, 'reports.html')
```

```
@login_required
def profile_view(request, user_id=None):
    from .models import Cadet, Attendance, Certificate, Event
    from django.contrib.auth.hashers import check_password
    from django.contrib import messages
    try:
        current_user_profile = UserProfile.objects.get(user=request.user)
    except UserProfile.DoesNotExist:
        current_user_profile = None

    # If user_id is provided, show that user's profile (with permission check)
    if user_id:
        # Only ANO can view other profiles
        if current_user_profile and current_user_profile.user_type != 'ano':
            return HttpResponseRedirect("Access denied. Only ANO can view other profiles.")

        try:
            target_user = User.objects.get(id=user_id)
            target_profile = UserProfile.objects.get(user=target_user)
        except (User.DoesNotExist, UserProfile.DoesNotExist):
            return HttpResponseRedirect("User not found.")

        cadet = Cadet.objects.filter(user=target_user).first()
        user = target_user
        user_profile = target_profile
        is_own_profile = False
    else:
        # Show current user's profile
```

```

cadet = Cadet.objects.filter(user=request.user).first()
user = request.user
user_profile = current_user_profile
is_own_profile = True

if request.method == 'POST' and is_own_profile:
    if 'delete_account' in request.POST:
        if user_profile and user_profile.user_type == 'cadet':
            from django.contrib.auth import logout
            # Capture the user instance first (request.user will become AnonymousUser after
logout)
            user_to_delete = request.user
            logout(request)
            try:
                # Delete the captured user instance
                user_to_delete.delete()
            except Exception:
                # If deletion fails, show an error
                messages.error(request, 'There was a problem deleting your account. Please contact
support.')
```

return redirect('profile')

```

                messages.success(request, 'Your account has been permanently deleted.')
                return redirect('home')
            else:
                messages.error(request, 'You are not authorized to delete this account.')
```

else:

```

    current_password = request.POST.get('current_password')
    new_password = request.POST.get('new_password')
    confirm_password = request.POST.get('confirm_password')

    if not check_password(current_password, request.user.password):
        messages.error(request, 'Current password is incorrect.')
    elif new_password != confirm_password:
        messages.error(request, 'New passwords do not match.')
    elif len(new_password) < 8:
        messages.error(request, 'New password must be at least 8 characters long.')
    else:
        request.user.set_password(new_password)
        request.user.save()
        messages.success(request, 'Password changed successfully.')
        return redirect('profile')

attendance_history = Attendance.objects.filter(cadet=cadet).order_by('-date') if cadet else []
certificates = Certificate.objects.filter(cadet=cadet) if cadet else []
        upcoming_events = Event.objects.filter(is_upcoming=True,
```

```
date__gte=date.today()).order_by('date')[:5]
```

```
context = {
    'user': user,
    'user_profile': user_profile,
    'cadet': cadet,
    'attendance_history': attendance_history,
    'certificates': certificates,
    'upcoming_events': upcoming_events,
    'is_own_profile': is_own_profile,
    'user_type': user_profile.user_type if user_profile else "",
}
return render(request, 'profile.html', context)
```

```
@login_required
```

```
def attendance_view(request):
    return render(request, 'attendance.html')
```

```
@login_required
```

```
def manage_events_view(request):
    from .models import Event
    try:
        user_profile = UserProfile.objects.get(user=request.user)
    except UserProfile.DoesNotExist:
        return HttpResponseForbidden("User profile not found.")

    # Allow both ANO and cadet to view events, but only ANO can manage (add/delete)
    if user_profile.user_type not in ['ano', 'cadet']:
        return HttpResponseForbidden("Access denied. Only ANO and Cadet can view events.")

    events = Event.objects.all().order_by('-date')
    return render(request, 'events_manage.html', {'events': events, 'user_type':
user_profile.user_type})
```

```
@login_required
```

```
def add_event_view(request: HttpRequest) -> HttpResponse:
    from .models import Event
    try:
        user_profile = UserProfile.objects.get(user=request.user)
    except UserProfile.DoesNotExist:
        return HttpResponseForbidden("User profile not found.")

    if user_profile.user_type != 'ano':
        return HttpResponseForbidden("Access denied. Only ANO can add events.")

    token = get_token(request)
```

```

if request.method == 'POST':
    title = request.POST.get('title')
    description = request.POST.get('description')
    date = request.POST.get('date')
    is_upcoming = request.POST.get('is_upcoming') == 'on'

```

```

Event.objects.create(
    title=title,
    description=description,
    date=date,
    is_upcoming=is_upcoming
)
return redirect('manage_events')

```

```

html = f'<!DOCTYPE html>
<html lang="en"><head><meta charset="UTF-8"><title>Add Event - NCC</title>
<style>
* {{ margin: 0; padding: 0; box-sizing: border-box; }}
body {{ font-family: 'Segoe UI', Arial, sans-serif; background: #f5f5f5; }}
.top-navbar {{ background: linear-gradient(135deg, #2c3e50 0%, #34495e 100%); color:
white; padding: 15px 30px; display: flex; justify-content: space-between; align-items: center; }}
.main-container {{ display: flex; min-height: calc(100vh - 70px); }}
.sidebar {{ width: 250px; background: white; box-shadow: 2px 0 10px rgba(0,0,0,0.1);
padding: 20px 0; }}
.sidebar-item {{ display: block; padding: 15px 25px; color: #555; text-decoration: none;
transition: all 0.3s; }}
.sidebar-item:hover {{ background: #f8f9fa; color: #2c3e50; }}
.content {{ flex: 1; padding: 30px; }}
.form-container {{ background: white; padding: 30px; border-radius: 10px; box-shadow: 0
2px 15px rgba(0,0,0,0.08); max-width: 600px; }}
.form-group {{ margin-bottom: 20px; }}
label {{ display: block; margin-bottom: 8px; font-weight: 600; color: #333; }}
input, textarea {{ width: 100%; padding: 12px; border: 1px solid #ddd; border-radius: 5px;
font-size: 16px; }}
.btn {{ padding: 12px 25px; background: #3498db; color: white; text-decoration: none;
border-radius: 5px; font-size: 16px; border: none; cursor: pointer; }}
.btn:hover {{ background: #2980b9; }}
.btn-success {{ background: #27ae60; }}
.btn-success:hover {{ background: #219a52; }}
</style></head>
<body>
<nav class="top-navbar"><div>🔔 NCC Management - Add Event</div><div><a
href="/dashboard/" style="color:white;margin-right:20px;">← Back to Dashboard</a><a
href="/logout/" style="color:white;">Logout</a></div></nav>
<div class="main-container">
<div class="sidebar">

```

```

<a href="/dashboard/" class="sidebar-item"><img alt="Dashboard icon" data-bbox="545 58 568 75"/> Dashboard</a>
<a href="/cadets/" class="sidebar-item"><img alt="Manage Cadets icon" data-bbox="515 82 538 99"/> Manage Cadets</a>
<a href="/attendance/" class="sidebar-item"><img alt="Attendance icon" data-bbox="548 103 571 120"/> Attendance</a>
<a href="/events/manage/" class="sidebar-item"><img alt="Manage Events icon" data-bbox="583 124 606 141"/> Manage Events</a>
<a href="/training/" class="sidebar-item"><img alt="Training Records icon" data-bbox="525 145 548 162"/> Training Records</a>
<a href="/reports/" class="sidebar-item"><img alt="Reports icon" data-bbox="518 166 541 183"/> Reports</a>
</div>
<div class="content">
<h1>Add New Event</h1>
<div class="form-container">
<form method="post" enctype="multipart/form-data">
<input type="hidden" name="csrfmiddlewaretoken" value="{token}">
<div class="form-group"><label>Title:</label><input type="text" name="title"
required></div>
<div class="form-group"><label>Description:</label><textarea
name="description" rows="4" required></textarea></div>
<div class="form-group"><label>Date:</label><input type="date" name="date"
required></div>
<div class="form-group"><label><input type="checkbox" name="is_upcoming"
checked> Upcoming Event</label></div>
<button type="submit" class="btn btn-success">Add Event</button>
</form>
</div>
</div>
</body></html>""
return HttpResponse(html)

```

```

@login_required
def edit_event_view(request, event_id):
    from .models import Event
    try:
        user_profile = UserProfile.objects.get(user=request.user)
    except UserProfile.DoesNotExist:
        return HttpResponseForbidden("User profile not found.")

    if user_profile.user_type != 'ano':
        return HttpResponseForbidden("Access denied. Only ANO can edit events.")

    event = get_object_or_404(Event, id=event_id)

    token = get_token(request)
    if request.method == 'POST':
        title = request.POST.get('title')
        description = request.POST.get('description')

```



```
date = request.POST.get('date')
is_upcoming = request.POST.get('is_upcoming') == 'on'
```

```
event.title = title
event.description = description
event.date = date
event.is_upcoming = is_upcoming
event.save()
```

```
return redirect('manage_events')
```

```
html = f'<!DOCTYPE html>
<html lang="en"><head><meta charset="UTF-8"><title>Edit Event - NCC</title>
<style>
* {{ margin: 0; padding: 0; box-sizing: border-box; }}
body {{ font-family: 'Segoe UI', Arial, sans-serif; background: #f5f5f5; }}
.top-navbar {{ background: linear-gradient(135deg, #2c3e50 0%, #34495e 100%); color:
white; padding: 15px 30px; display: flex; justify-content: space-between; align-items: center; }}
.main-container {{ display: flex; min-height: calc(100vh - 70px); }}
.sidebar {{ width: 250px; background: white; box-shadow: 2px 0 10px rgba(0,0,0,0.1);
padding: 20px 0; }}
.sidebar-item {{ display: block; padding: 15px 25px; color: #555; text-decoration: none;
transition: all 0.3s; }}
.sidebar-item:hover {{ background: #f8f9fa; color: #2c3e50; }}
.content {{ flex: 1; padding: 30px; }}
.form-container {{ background: white; padding: 30px; border-radius: 10px; box-shadow: 0
2px 15px rgba(0,0,0,0.08); max-width: 600px; }}
.form-group {{ margin-bottom: 20px; }}
label {{ display: block; margin-bottom: 8px; font-weight: 600; color: #333; }}
input, textarea {{ width: 100%; padding: 12px; border: 1px solid #ddd; border-radius: 5px;
font-size: 16px; }}
.btn {{ padding: 12px 25px; background: #3498db; color: white; text-decoration: none;
border-radius: 5px; font-size: 16px; border: none; cursor: pointer; }}
.btn:hover {{ background: #2980b9; }}
.btn-success {{ background: #27ae60; }}
.btn-success:hover {{ background: #219a52; }}
</style></head>
<body>
<nav class="top-navbar"><div>🏰 NCC Management - Edit Event</div><div><a
href="/dashboard/" style="color:white;margin-right:20px;">← Back to Dashboard</a><a
href="/logout/" style="color:white;">Logout</a></div></nav>
<div class="main-container">
<div class="sidebar">
<a href="/dashboard/" class="sidebar-item">🏠 Dashboard</a>
<a href="/cadets/" class="sidebar-item">👥 Manage Cadets</a>
```

```

<a href="/attendance/" class="sidebar-item">📅 Attendance</a>
<a href="/events/manage/" class="sidebar-item">📅 Manage Events</a>
<a href="/training/" class="sidebar-item">🔄 Training Records</a>
<a href="/reports/" class="sidebar-item">📊 Reports</a>
</div>
<div class="content">
<h1>Edit Event</h1>
<div class="form-container">
<form method="post" enctype="multipart/form-data">
<input type="hidden" name="csrfmiddlewaretoken" value="{token}">
<div class="form-group"><label>Title:</label><input type="text" name="title"
value="{event.title}" required></div>
<div class="form-group"><label>Description:</label><textarea name="description"
rows="4" required>{event.description}</textarea></div>
<div class="form-group"><label>Date:</label><input type="date" name="date"
value="{event.date}" required></div>
<div class="form-group"><label><input type="checkbox" name="is_upcoming"
{"checked" if event.is_upcoming else ""}> Upcoming Event</label></div>
<button type="submit" class="btn btn-success">Save Changes</button>
</form>
</div>
</div>
</body></html>""""
return HttpResponse(html)

```

@login_required

def delete_event_view(request: HttpRequest, event_id: int) -> HttpResponse:

if request.method != 'POST':

return HttpResponseNotAllowed(['POST'])

try:

user_profile = UserProfile.objects.get(user=request.user)

except UserProfile.DoesNotExist:

return HttpResponseForbidden("User profile not found.")

if user_profile.user_type != 'ano':

return HttpResponseForbidden("Access denied. Only ANO can delete events.")

event_to_delete = get_object_or_404(Event, id=event_id)

event_to_delete.delete()

return redirect('manage_events')

@login_required

def manage_contact_view(request):

```

from .models import ContactFeedback
try:
    user_profile = UserProfile.objects.get(user=request.user)
except UserProfile.DoesNotExist:
    return HttpResponseForbidden("User profile not found.")

if user_profile.user_type != 'ano':
    return HttpResponseForbidden("Access denied. Only ANO can manage contact
feedbacks.")

feedbacks = ContactFeedback.objects.all().order_by('-timestamp')
    return render(request, 'manage_contact.html', {'feedbacks': feedbacks, 'user_type':
user_profile.user_type})

@login_required
def delete_contact_view(request: HttpRequest, feedback_id: int) -> HttpResponse:
    if request.method != 'POST':
        return HttpResponseNotAllowed(['POST'])
    try:
        user_profile = UserProfile.objects.get(user=request.user)
    except UserProfile.DoesNotExist:
        return HttpResponseForbidden("User profile not found.")

    if user_profile.user_type != 'ano':
        return HttpResponseForbidden("Access denied. Only ANO can delete contact feedbacks.")

    feedback_to_delete = get_object_or_404(ContactFeedback, id=feedback_id)
    feedback_to_delete.delete()

    return redirect('manage_contact')

```

6.3 Testing Approach

For the NCC Unit Management Website, we adopted a comprehensive and multi-dimensional testing strategy, including Unit Testing, Usability Testing, and Security Testing, to ensure the application's reliability, efficiency, and robustness.

6.4 Unit Testing

Unit Testing involved rigorously testing individual components of the website's modules, such as Cadet Registration, Attendance Management, Training Records, and Reports. Each function, view, and model was tested in isolation, confirming their correctness and handling of expected inputs. This allowed early detection of defects, ensuring that each building block of the application performs as intended before integrating into the whole system. We used Django's built-in testing framework to write automated tests that validate core functionalities and maintain system stability during ongoing development.

6.5 Usability Testing

Usability Testing was performed with real users interacting with the website to evaluate the overall user experience. Feedback was collected on ease of navigation across pages like Dashboard, Manage Cadets, and Profile sections. The level of intuitiveness, clarity of instructions, and responsiveness of interfaces were assessed to identify pain points. This testing helped us refine the user interface to make the website straightforward and satisfying to use, ensuring that even users with limited technical knowledge could manage cadet information efficiently.

6.6 Security Testing

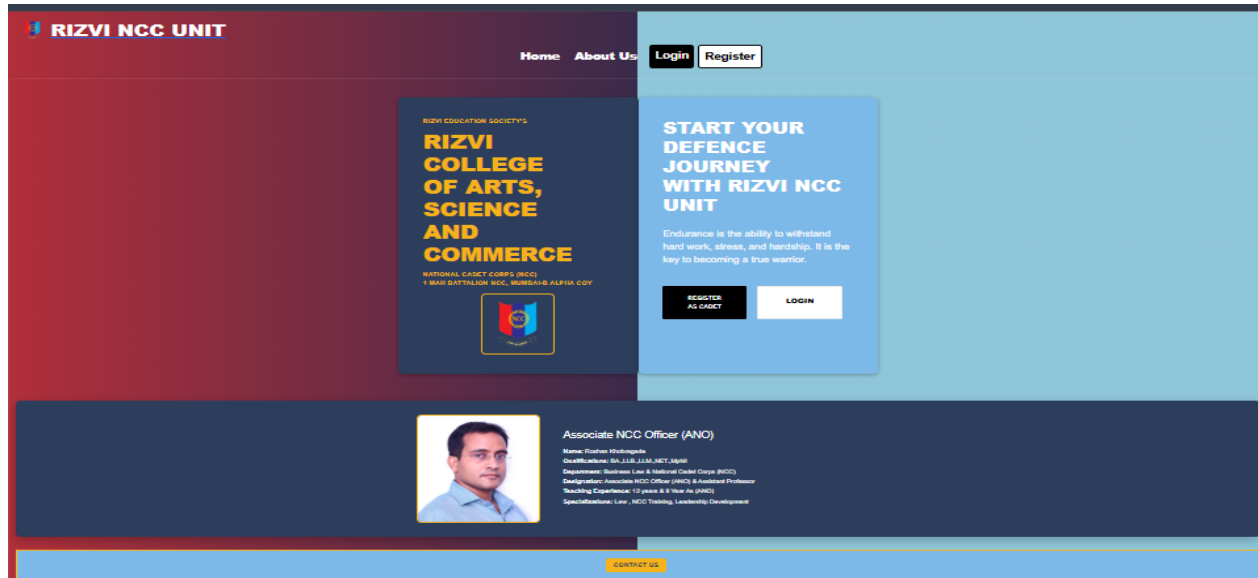
Security Testing was a critical part of our project implementation to safeguard sensitive cadet data and user credentials. The website was systematically checked for common vulnerabilities such as cross-site scripting (XSS), SQL injection, and unauthorized access. Tests included penetration testing and code reviews to verify encryption and secure authentication mechanisms. Our objective was to uphold the confidentiality, integrity, and availability of data, ensuring a secure platform for all NCC unit activities and preventing potential security breaches that could compromise user trust.

6.7 Test Cases

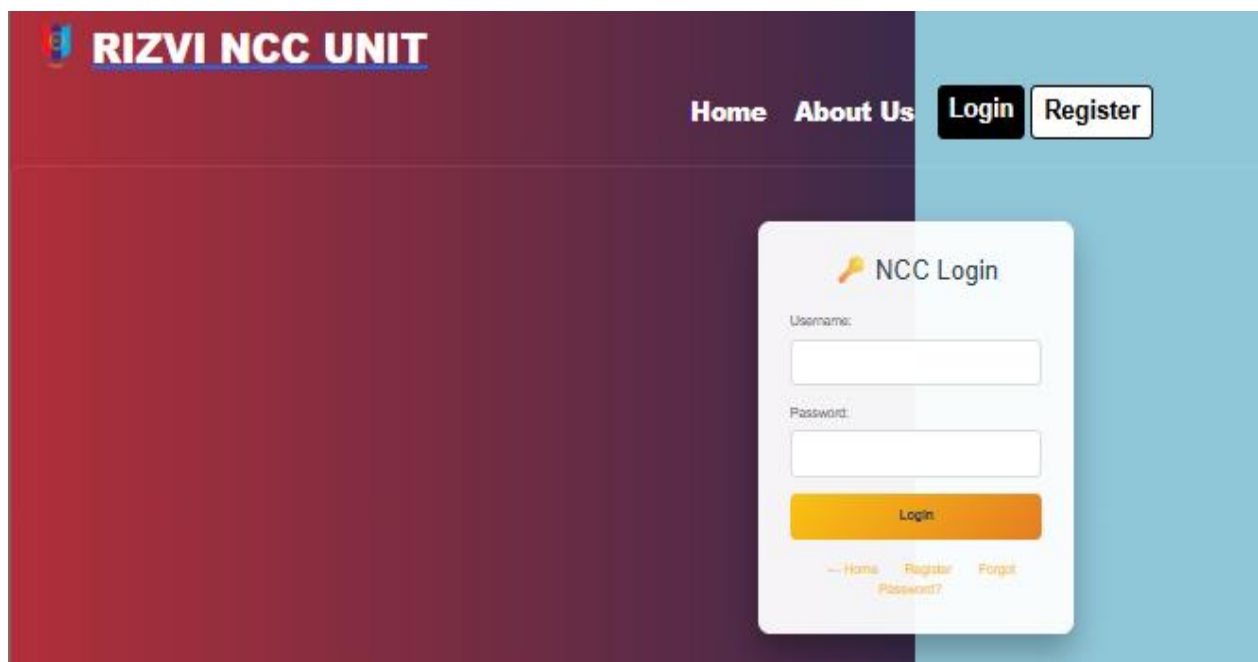
Test-Case ID	Case	Action	Expected Results	Actual Results	Status
1	Cadet Registration	Enter cadet details and submit form	Cadet successfully registered	Cadet successfully registered	Pass
2	Login	Login with valid credentials	Homepage/dashboard loads for user	Homepage/dashboard loads for user	Pass
3	Attendance Marking	Mark attendance for registered cadet	Attendance status updated in database	Attendance status updated in database	Pass
4	Training Record Entry	Add new training record to cadet profile	Training record appears in cadet's profile	Training record appears in cadet's profile	Pass
5	Certificate Download	Download certificate for eligible cadet	Certificate PDF gets downloaded successfully	Certificate PDF gets downloaded successfully	Pass
6	Manage Events	Create new event and save	Event appears in event list	Event appears in event list	Pass
7	Contact Message Form	Submit message via contact form	Message stored and confirmation shown	Message stored and confirmation shown	Pass
8	Profile Update	Edit profile and save changes	Updated details show in profile	Updated details show in profile	Pass

CHAPTER 7. RESULTS

Homepage



Login page



Registration Page

RIZVI NCC UNIT

Home

About Us

Login

Register

NCC Registration

Username

Full name

Last name

Email

Phone

Reg no

Rank

Cadet

Date joined

Blood group

Image

Choose file

No file chosen

Password

Password confirmation

REGISTER

< Home

Login

ANO Dashboard

Dashboard

Manage Cadets

Attendance

Fallin Records

Certificates

Reports

Event Manage

Contact Messages

My Profile

Back to Website

Welcome, ara123 / AND

Logout

Dashboard Overview

4
Total Cadets

4
Active Cadets

0
Training Programs

1
Upcoming Events

Quick Actions

My Profile

View and manage your personal profile

View Profile

Cadets Profile

view all cadets profile of RIZZI NCC UNIT

Manage Cadets

Take Attendance

Mark today's attendance for training sessions

Take Attendance

FALLIN SCHEDULE

view fallin schedule

View Training

EVENTS

view new upcoming events

Manage Events

Cadet Reports

View and manage cadet activity reports

View Reports

Contact Messages

View and manage messages submitted via Contact Us form

View Messages

Registered Cadets

Username	Name	Reg No	Phone	Actions
jay789	JAY GUPTA	MH23CA012345	1234567890	Edit Delete
harsha789	HARSHA	MH23SC0456789	0123456789	Edit Delete
cadet789	cadet	MH23SC0456782	0123456789	Edit Delete
harshali407	Harshali	MH23SC0456785	9876543210	Edit Delete

Attendance Page

Dashboard

Manage Cadets

Attendance

Fallin Records

Certificates

Reports

Event Manage

Contact Messages

My Profile

Back to Website

Rizvi NCC Unit

Welcome, ano123 | ANO

Logout

Attendance Management

Select Date: 07-10-2025

LOAD ATTENDANCE

VIEW SUMMARY

Registration No.	Name	Rank	Company	Status
MHSDA012345	JAY GUPTA	Cadet	A Company	☐ Present ☐ Absent ☐ In Camp
MH23SDA56789	HAMZA	Cadet	A Company	☐ Present ☐ Absent ☐ In Camp
MH23SDA56782	cadet	Cadet	A Company	☐ Present ☐ Absent ☐ In Camp
MH23SDA56785	Hanzala	Cadet	A Company	☐ Present ☐ Absent ☐ In Camp

SAVE ATTENDANCE

Fallin Schedule Page

Dashboard

Manage Cadets

Attendance

Fallin Records

Certificates

Reports

Event Manage

Contact Messages

My Profile

Back to Website

Rizvi NCC Unit

Welcome, ano123 | ANO

Logout

Training Record

Manage Training Schedule

dd-mm-yyyy

--:--

Activity

Location

Add Another Entry

Save Schedule

Training Record for 10/04/2025 (Saturday)

Time	Activity	Location	Action
7 a.m.	Reporting	Parade Ground	Delete
7:30 a.m.	PT	PT Ground	Delete
10 a.m.	Drill	Parade Ground	Delete

Contact Messages Send by Users Page

Dashboard

Manage Cadets

Attendance

Fallin Records

Certificates

Reports

Event Manage

Contact Messages

My Profile

Back to Website

Rizvi NCC Unit

Welcome, ano123 | ANO

Logout

Contact Messages

Name: Hanzala Sayyed
Email: abc@gmail.com
Subject: hello
Message: hello sir
Date: 08/10/2025
Time: 01:09 PM

Delete

Name: juned
Email: juned@gmail.com
Subject: hello
Message: how are you ?
Date: 05/10/2025
Time: 11:51 AM

Delete

Name: Hanzala Sayyed
Email: abc@gmail.com
Subject: hello
Message: hello world
Date: 05/10/2025
Time: 06:39 AM

Delete

Name: Hanzala Sayyed
Email: abc@gmail.com
Subject: hello
Message: hello world
Date: 04/10/2025
Time: 03:36 AM

Delete

Name: Hanzala Sayyed
Email: abc@gmail.com
Subject: hello
Message: hello world
Date: 04/10/2025
Time: 03:33 AM

Delete

Cadets Dashboard

Dashboard

Cadets Profile

Attendance

Fallin Records

Certificates

Reports

Events

Contact Us

My Profile

Back to Website

Rizvi NCC Unit

Welcome, hamza786 | CADET

Logout

Quick Actions



My Profile
View and manage your personal profile

View Profile



Cadets Profile
view all cadets profile of RIZVI NCC UNIT

Manage Cadets



FALLIN SCHEDULE
view fallin schedule

View Training



EVENTS
view new upcoming events

Manage Events



Cadet Reports
View and manage cadet activity reports

View Reports

Contact us Page

Dashboard

Cadets Profile

Attendance

Fallin Records

Certificates

Reports

Events

Contact Us

My Profile

Back to Website

Rizvi NCC Unit

Welcome, hamza786 | CADET

 Logout

Contact Us

Send Message

Cadets Profile

Dashboard

Cadets Profile

Attendance

Fallin Records

Certificates

Reports

Events

Contact Us

My Profile

Back to Website

Rizvi NCC Unit

Welcome, hamza786 | CADET

 Logout

Cadets Profiles



Name: HAMZA
Reg.no: MH23SDA56789
Rank: Cadet
Ph.No: 0123456789
Blood Group: B-



Name: Hanzala
Reg.no: MH23SDA56785
Rank: Cadet
Ph.No: 9930525095
Blood Group: O+



Name: JAY GUPTA
Reg.no: MHSDA012345
Rank: Cadet
Ph.No: 1234567890
Blood Group: B+



Name: cadet
Reg.no: MH23SDA56782
Rank: Cadet
Ph.No: 0123456789
Blood Group: B-

CHAPTER 8. CONCLUSION AND FUTURE SCOPE

8.1 Conclusion

.The NCC Unit Management Website represents a major advancement in automating and streamlining NCC-related administrative and academic functions. By digitizing core processes such as cadet registration, attendance, training records, event management, and reporting, the platform delivers significant improvements in both efficiency and accuracy for unit managers, instructors, and cadets. The intuitive, web-based interface ensures that users can easily access all important features, reducing paperwork and manual errors. The system's secure authentication mechanisms mean sensitive cadet and faculty data remains protected at all times, while real-time updates and reporting tools aid decision-making and enhance transparency within the unit. Overall, this digital solution modernizes NCC record-keeping and management, creates a user-friendly experience for all stakeholders, and lays a solid foundation for scalable growth

8.2 Future Scope and Enhancement

The successful development and deployment of the NCC Unit Management Website highlights tremendous opportunities for ongoing innovation and growth. As this digital platform matures, several promising enhancements and technological advancements can be envisioned that will extend its utility and impact for NCC units, administrators, cadets, and educational institutions nationwide.

One major direction for future work is the integration of advanced data analytics and dashboard features. By tracking engagement, event participation, attendance trends, and performance statistics, the website can deliver actionable insights to both cadet leaders and instructors. The use of visual charts and automated reports will assist officers in making faster, more informed decisions, contributing to even greater transparency and accountability.

To improve accessibility and usability, upcoming versions may feature a dedicated mobile app or progressive web app (PWA), enabling users to register, check attendance, and view unit news from any device. Additionally, features like instant SMS/email notifications for announcements, drill calls, or certificate availability could further enhance communication and cohesion within the unit community.

Security and privacy will remain a top priority, with future enhancements aimed at

implementing more robust role-based access control, multi-factor authentication, and encrypted messaging between users. Integrating Single Sign-On (SSO) for institutional logins will support easier and more secure access for cadets and staff.

The system can also evolve by supporting seamless integration with government and university databases for direct eligibility verification, and by providing modules for digital certificate issuance, biometric attendance, and automated event scheduling and reminders. Expansion possibilities include modules for online training, resources for self-study, and video-based virtual drill practice.

A feedback and suggestion portal could also be added to the platform, ensuring that the voices of cadets and officers directly guide future upgrades. Support for multi-language interfaces will allow wider participation across India and beyond, ensuring inclusivity.

In the long term, by maintaining a modular design and open API standards, the NCC Unit Management Website can become part of a larger ecosystem of educational tools, supporting not only operational efficiency but also leadership development, career tracking, and alumni engagement. Emphasizing scalability, adaptability, and strong data protection, this project is positioned to serve as a benchmark for digital transformation in institutional management and youth leadership development for years to come.

CHAPTER 9. REFERENCES

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